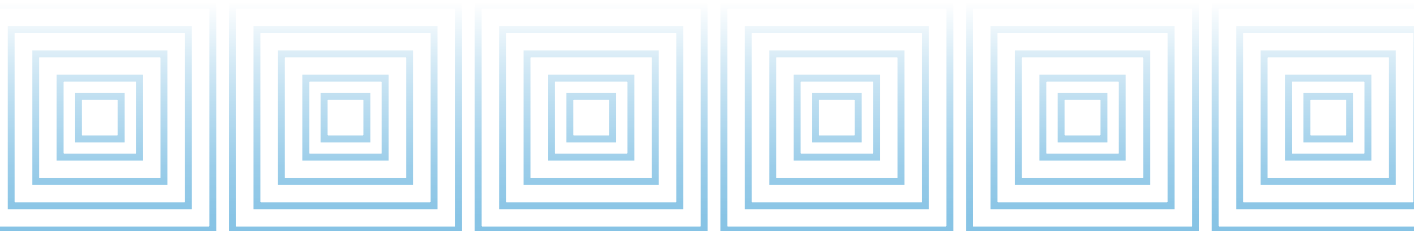


PHYSICS SYLLABUS

ADVANCED LEVEL (S4 - S6)

Kigali, 2026



Experimental Version

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FOREWORD

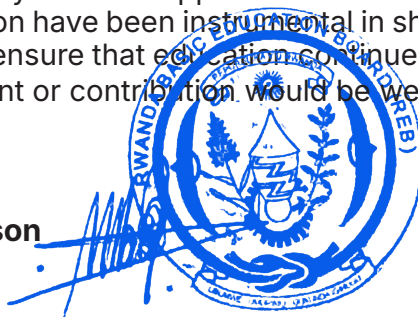
The Rwanda Basic Education Board (REB) is honoured to present the advanced level Physics syllabus (S1–S3). It serves as an official guide for competence-based teaching and learning, ensuring consistency in delivering quality education across the country. Developed collaboratively by subject specialists, educators, and curriculum developers, it supports the implementation of the Competence-Based Curriculum (CBC) and reflects Rwanda's commitment to nurturing learners' knowledge, skills, values, and attitudes.

Quality education is the foundation of Rwanda's sustainable development and prosperity. Aligned with the national education philosophy, this syllabus promotes critical thinking, creativity, inclusiveness, and holistic learner development. It aims to empower young people with the competences necessary to achieve full potential, equipping them with relevant knowledge, skills, the right attitudes and values to integrate effectively into society and pursue opportunities in a competitive world.

Anchored in Rwanda Vision 2050 and the National Strategy for Transformation (NST2), this syllabus supports the nation's goal of building a knowledge-based economy driven by innovation, resilience, and global competitiveness. It prioritises learner-centred pedagogy, active learning, and the practical application of knowledge to prepare students not only for academic success but also to become responsible, ethical, and productive citizens who can contribute to Rwanda's continued transformation. This syllabus reflects our determination to equip learners with 21st-century skills, foster innovation, and instil values of integrity, social responsibility, and lifelong learning. Its success depends largely on the dedication and professionalism of teachers and school leaders. We therefore call upon all educators to embrace the approaches, methodologies, and assessment strategies outlined herein with commitment and passion, transforming classrooms into dynamic spaces of curiosity, creativity, and inspiration.

On behalf of REB, I extend my heartfelt appreciation to all who contributed to the development of this syllabus. Your expertise and dedication have been instrumental in shaping a curriculum that responds to the needs of our learners. Together, we can ensure that education continues to be a powerful driver of our nation's development and prosperity. Any comment or contribution would be welcome to the improvement of this syllabus.

Dr. MBARUSHIMANA Nelson
Director General, REB



ACKNOWLEDGEMENT

The development of this syllabus has been a collective effort that reflects the dedication, expertise, and commitment of many stakeholders. On behalf of the Curriculum, Teaching and Learning Resources Department of the Rwanda Basic Education Board (REB), I wish to express my deepest appreciation to all who contributed to the successful preparation of this Advanced Level Physics syllabus (S4–S6).

Special thanks go to the subject experts, curriculum developers, teachers, and education partners whose insights, feedback, and professional contributions have been invaluable in shaping this syllabus. Your collaboration has ensured that the syllabus is not only aligned with Rwanda's Competence-Based Curriculum but also responsive to the aspirations of Vision 2050 and the National Strategy for Transformation (NST2).

We also acknowledge the invaluable support of development partners and education stakeholders who have consistently provided technical assistance in curriculum adaptation. Your partnership has greatly contributed to strengthening the quality of Rwanda's education system.

Finally, we call upon teachers, school leaders, and all education practitioners to use this syllabus with commitment and professionalism, ensuring that classrooms become vibrant learning environments that nurture knowledge, skills, values, and positive attitudes among learners. Through your dedication, this syllabus will fulfil its purpose of empowering learners to realize their full potential and contribute meaningfully to Rwanda's sustainable development.



MURUNGI Joan

**Head of Curriculum, Teaching and Learning Resources Department
Rwanda Basic Education Board (REB)**

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1. GENERAL INTRODUCTION

1.1. Background to the syllabus adaptation

The adaptation of the Physics syllabus for upper secondary school reflects Rwanda's ongoing commitment to strengthening the Competence-Based Curriculum (CBC), first rolled out in 2015. Continuous feedback from teachers, school leaders, and education partners highlighted the need for improvements in content delivery, time allocation, and responsiveness to learner needs.

This second edition incorporates the Ministerial directive to harmonize school hours to 33 weeks of learning annually and ensures greater coherence, inclusiveness, and alignment with national priorities such as *Vision 2050* and the *National Strategy for Transformation (NST2)*, which emphasize a knowledge-based economy, responsible citizenship, and sustainable development.

Physics at the upper secondary level aims to equip learners with foundational knowledge, practical skills, and positive values needed to identify opportunities, solve real-life problems, and contribute meaningfully to their communities. The subject encourages learners to think creatively, act ethically, and develop a sense of initiative, self-confidence, and social responsibility.

1.2. Rationale of teaching and learning Physics

Physics helps learners understand natural phenomena and the physical laws that govern the world around them. It develops scientific inquiry, critical thinking, and problem-solving skills through observation, experimentation, and application.

By learning Physics, students gain practical knowledge useful in everyday life, such as understanding electricity, motion, heat, and energy. It also prepares them for further studies and careers in science, technology, and engineering, and promotes creativity, innovation, and responsible use of natural resources for sustainable national development.

1.2.1. Physics and society

Physics is one of the natural science subjects and contributes significantly to global socioeconomic transformation through its discoveries. These have led to development of new technologies in all fields of production and are beneficial to mankind. Applications of Physics knowledge is evident in industries such as manufacturing, transport (automobiles, trains, planes, etc.), medicine and Information and Communication Technology (ICT).

Physics significantly contributes to the advancement of new technologies that arise from theoretical breakthroughs. For example, advances and understanding of electromagnetism or nuclear physics has led to the development of new products, which have dramatically transformed the modern society. Discoveries in physics have resulted in innovations such as televisions, computers, electrical appliances, and nuclear energy systems, as well as advancements in thermodynamics and mechanics that fueled industrialization.

Furthermore, physics supports Rwanda's vision of building a knowledge-based economy by promoting science and technology. It equips learners with essential skills and competencies needed to compete effectively in both regional and global job markets.

1.2.2. Physics and learner

Physics is a worthwhile subject because it prepares students for the real world of work by providing career pathways in mechanical engineering, construction engineering, information and communication technology and other related fields. Physics provides skills that guide in the construction of theories and laws that help to explain natural phenomenon and enable management of the environment.

It also provides answers to problems faced in our modern society by empowering students to be creative and innovative leading to independent approaches of solving daily life problems. Through physics, learners explore the laws and rules that govern all natural phenomena observed in the universe.

1.2.3. Competences

A competence is the ability to use an appropriate combination of knowledge, skills, attitudes, values, and behaviours to accomplish a task successfully. In this syllabus, competences reflect a holistic blend of these elements, enabling learners to apply them confidently and appropriately in diverse real-life situations.

By acquiring these competences, learners become better equipped to handle challenges, make informed decisions, act responsibly, and contribute positively as active members of society. The competences are embedded in the broad subject competences, yearly objectives, and unit-specific learning competences. Both generic competences and basic competences must be emphasized and consistently reflected in the teaching and learning process.

Teachers are encouraged to design learning experiences that engage learners in meaningful tasks, allowing them to practice and demonstrate competences both during and at the end of the learning process.

1.2.4. Generic competences

Generic competences are cross-cutting abilities developed across all subjects. They equip learners with essential lifelong learning skills and underpin all other competences in the curriculum.

- **Critical thinking and problem-solving skills:** The acquisition of such skills will help learners to think deeply, analyze situations, and make reasoned judgments to find effective solutions in real-life situations.
- **Creativity and innovation:** The acquisition of such skills will help learners to take the initiative and use imagination beyond the knowledge provided, to generate new ideas and construct new concepts.
- **Research:** This will help learners acquire the ability to gather, analyze, and interpret information from a variety of sources, including observations, interviews, and case studies to inform decisions and develop evidence-based conclusions.
- **Communication in official languages:** Teachers, irrespective of their subject, will ensure proper use of the language of instruction to help learners communicate clearly, confidently, and respectfully in both spoken and written forms, enabling them to express ideas, negotiate, and present information effectively for logical and coherent exchange.

- **Cooperation, interpersonal management and life skills:** This competence enables learners to work effectively with others, respect diverse opinions, and show empathy in group activities. It fosters moral values, responsibility, and active participation in community-oriented tasks such as environmental protection, health promotion, and service. Learners also build social skills that enhance teamwork, leadership, and adaptability in addressing life's challenges.
- **Lifelong learning:** Learners cultivate curiosity, initiative, and adaptability by linking classroom knowledge to real-life experiences. This empowers them with the ability to update knowledge and skills independently, cope with evolving societal needs, and pursue personal growth and continuous development.

1.2.5. Physics broad competences

At the end of upper secondary education, learners should be able to demonstrate the following broad competences specific to Physics:

- Analyze and explain physics phenomena relating to life experience.
- Use and experiment with a range of scientific and technological tools and equipment and draw appropriate conclusions.
- Demonstrate curiosity, research skills and creativity.
- Apply scientific inquiry and methods to investigations.
- Apply knowledge of mathematics and technology to scientific investigation.
- Observe, analyse, evaluate, and interpret without prejudice and make reasonable decisions.
- Use principles of scientific methods and experimental techniques to solve specific problems in life.
- Develop attitudes in which scientific investigation depends on honesty, persistence, critical thinking and tolerance of uncertainty.
- Appreciate the scientific, social, economic, environmental and technological implications of physics.

- Identify safe and appropriate techniques used in the preparation, storage, dispensing and supervision of materials used in science instructions.
- Identify national legal requirements and standards for safe preparation, use, storage, and disposal of the materials used.

1.2.6. Physics and developing competences

The policy documents based on national aspirations identify some 'Basic Competences' alongside the 'Generic Competences' that will develop higher order thinking skills and help students to learn subject content and promote application of acquired knowledge and skills.

Through observation, experimentation, and presentation of information during the learning process, the learner will not only develop deductive and inductive skills, but also acquire cooperation, communication, critical thinking and problem-solving skills. This will be realized when learners make presentations leading to inferences and conclusions at the end of learning unit. This will be achieved through learner group work and cooperative learning of physics, which in turn will promote interpersonal relations and teamwork.

The manipulation of apparatus and data during class experiments and the undertaking of project work by learners will involve analytical and problem-solving skills directed towards innovation, creativity and research activities by the learners.

The acquired knowledge from learning physics should help develop a responsible citizen who uses scientific reasoning and attitudes to develop confidence in reasoning independently. The learner should show concern for individual attitudes, environmental protection and comply with the scientific method of reasoning. The scientific method should be applied with the necessary rigor and intellectual honesty to promote critical thinking while systematically pursuing the line of thought.

2. PEDAGOGICAL APPROACH

Learners enjoy learning when they are actively involved in the learning process with a high degree of participation, contribution and presentation. At the same time, each learner is an individual with their own needs, pace of learning, experiences and abilities. Teaching strategies must therefore be varied and flexible within well-structured sequences of lessons. Learner-centered education does not mean that the teacher is no longer responsible for learner's learning.

2.1. Role of the Teacher

The Competence-Based Curriculum (CBC) aims to make learning active, enjoyable, and meaningful by focusing on the development of skills, attitudes, and understanding rather than rote memorization. Teachers are expected to move from traditional teaching methods to facilitation, where learners take an active role in discovering and constructing knowledge. This involves identifying learners' individual needs, guiding them through inquiry-based activities, and encouraging participation through discussions, experiments, and problem-solving tasks.

Learning under CBC is participatory and cooperative, allowing learners to work individually, in pairs, or in groups to explore and apply concepts. Teachers should use a variety of teaching and learning resources such as models, laboratory materials, charts, simulations, ICT tools, and textbooks to enrich the learning process. They also help learners develop research and note-taking skills, enabling them to use different resources effectively and become independent, confident, and competent learners.

2.2. Role of the Learner

The teaching and learning process activities are tailored towards creating a learner friendly environment based on the abilities, needs, experiences and interests of the learner.

The learning activities are organized in such a way that they encourage learners to construct their own knowledge (minds-on and hands-on activities) either individually or in groups. The learner should suggest how to solve challenging problems exposed to them.

In practical lessons, learners will work in groups depending on the availability of the apparatus, however if apparatus numbers permit, then they work individually. Working on simple project work individually will be encouraged and emphasized. Learners should use textbooks and other resources for complementing the knowledge acquired in classroom. Learners should strive to become thinkers, inquirers, problem solvers, and communicators. Learners should be principled, open-minded, caring, risk takers, and balanced in reflection.

2.3. Inclusive and Special Needs Education

In line with Rwanda's inclusive education policy, all learners regardless of ability have the right to access quality education. Teachers adapt teaching methods, materials, and assessment strategies to meet diverse needs, including those of learners with disabilities. Differentiated instruction, assistive technologies, peer support, and individualized approaches are used to ensure equitable participation and achievement of learning outcomes. This inclusive approach recognizes learners' varied paces, experiences, and abilities, creating an environment where every learner can thrive.

3. ASSESSMENT APPROACH

Assessment is the process of evaluating teaching and learning processes, by collecting and interpreting evidence of an individual learner's progress and to make a judgment about a learner's achievements measured against defined standards, providing constructive feedback to guide both teaching and learning.

Assessment is an integral part of the teaching learning process. In the new competence-based curriculum, assessment must also be competence-based, where a learner is given a complex situation related to their everyday life and asked to try to overcome the situation by applying what they have learned.

Assessment in physics focuses on evaluating learners' understanding of scientific concepts, their ability to apply these concepts to real-life situations, and their practical skills in conducting experiments. It also measures learners' scientific inquiry, problem-solving, and critical thinking abilities, as well as their attitudes toward safety, teamwork, and environmental responsibility in scientific work.

Assessment should therefore be comprehensive, covering both formative (continuous) and summative (end-of-term or national) evaluations. It should be transparent, fair, inclusive, and competence aligned.

3.1. Types of assessment

3.1.1. Diagnostic Assessment

Diagnostic Assessment is carried out before the teaching and learning process begins to determine learners' existing knowledge, skills, attitudes, and misconceptions about a topic. It helps the teacher identify learners' strengths and weaknesses so that lessons can be tailored to meet their individual needs. In physics, diagnostic assessment may include short quizzes, brainstorming sessions, discussions, or concept maps to find out what learners already know about concepts. The information gathered guides the teacher in planning appropriate instructional strategies and providing the necessary support to enhance effective learning.

3.1.2. Formative and continuous assessment (Assessment for learning)

Formative and Continuous Assessment are ongoing processes used to monitor learners' progress and provide feedback throughout the learning period. They help teachers identify learners' strengths, weaknesses, and learning needs, allowing timely support and improvement. In physics, these assessments may include class exercises, practical activities, quizzes, observations, oral questioning, and projects. The teacher continuously checks how well learners are mastering both subject content and key competences, ensuring steady progress and positive behaviour change before moving to the next unit.

3.1.3. Summative assessment (Assessment of learning)

When assessment is used to judge a learner's competence or performance, it serves a **summative purpose**. Summative assessment provides a clear picture of the learner's achievement at a specific point in time. Its main goal is to determine whether learning objectives have been met and to use the results for grading, ranking,

progression, selection, or certification. It requires learners to demonstrate mastery of all expected competences in an integrated manner.

Summative assessments are conducted at different levels:

End of Term I: A school-based assessment administered after instruction to evaluate learners' mastery of Term I content.

End of Term II: A district-coordinated assessment measuring Term II learning outcomes, developed by schools and validated at the district level.

End of Term III: A national assessment prepared and moderated to evaluate mastery of the entire year's content.

National Examinations: High-stakes national assessments administered at the end of key education cycles (P6, S3, S6) for cycle transition and certification. These exams integrate objective, structured, and applied tasks aligned with **Competence-Based Curriculum (CBC)** principles.

In Advanced Level, all students will take 11 subjects in total -7 core subjects and 4 compulsory subjects. The compulsory subjects include ICT, Entrepreneurship, General and Religious Studies, and Physical Education and Sport, which are mandatory for all learners across all pathways. All subjects will be assessed at the national level, except for 4 compulsory subjects which will be assessed at the school level only.

Finally, average scores from school, district, and national summative assessments are weighted and included in the final national examination grading starting with a 10% contribution, which will gradually increase as teachers gain more experience and confidence in assessment practices.

3.2. Record keeping

Record keeping is the gathering facts and evidence from assessment instruments and using them to judge the student's performance by assigning an indicator against the set criteria or standard. Whatever assessment procedures used shall generate data in the form of scores, which will be carefully recorded and stored in a

portfolio. The scores will contribute to remedial actions, alternative instructional strategies, feedback to the learner, for parents to check the learning progress and for teachers to advise accordingly, as well as to the final assessment of the students.

This portfolio is a folder (or binder or even a digital collection) containing the student's work, as well as the student's evaluation of the strengths and weaknesses of the work. Portfolios reflect not only work produced (such as papers and assignments), but also it is a record of the activities undertaken over time as part of student learning including the worksheets. The portfolio output (formative assessment) will be considered only as enough for three years of a level. It will serve as a verification tool for each learner that they attended the whole learning before they undergo the summative assessment for the subject. The results from the portfolio will contribute 50% of summative assessment of each year.

3.3. Item writing in summative assessment

Before developing a question paper, a plan, or specification of what is to be tested or examined, it must be elaborated to show the units or topics to be tested on, the number of questions in each level of Bloom's taxonomy and the marks allocation for each question. In a competence-based curriculum, questions from higher levels of Bloom's taxonomy should be given more weight than those from knowledge and comprehension levels.

Before developing a question paper, the item writer must ensure that the test or examination questions are tailored towards competence-based assessment by doing the following:

- Identify topic areas to be tested from the subject syllabus.

- Outline subject matter content to be considered as the basis for the test.

- Identify learning outcomes to be measured by the test.

- Preparing a test blueprint or table of specifications.

- Using verbs that require higher order thinking beyond memorization.

Applying transparent and consistent marking schemes.

Greater weight is given to higher-order cognitive skills as stated in the syllabus.

3.4. Structure and format of the examinations

There will be 3 papers in the Physics Subject to be examined. Time allocated for all papers will depend on their respective weight and in special case, it can depend on the needs of the learners. The papers will be structured as follows;

- Structured answer questions consist of a variable number of questions, of variable mark value. All questions will be based on the A` Level syllabus content. Candidates will answer all questions and on the question paper.
- Unstructured answer questions or extended paper in which all questions will be based on the A` level syllabus, but may require knowledge of material first encountered in the previous level of study.
- Practical paper will consist of experiments drawn from different areas of the syllabus that require candidates to carry out practical work in timed conditions. Candidates may not enter for single papers on the first occasion or for resit purposes. All components will be externally assessed.

Component Weighting

PAPER	COMPONENT	WEIGHT
Paper 1	- Measures knowledge and understanding (lower order thinking level). - Structured short answer questions. - All questions will be based on the syllabus content.	Structured short answer questions will have 30% of the final marking of the assessment.
Paper 2	- Consists of question with number of parts each with a variable mark value, which measures skills and advanced level of understanding (higher order thinking level). - Unstructured answer questions or extended essay questions. All questions will be based on the syllabus but may require knowledge of material first encountered in the previous syllabus of the same subject.	Unstructured answer questions will have 40 % of the final marking of the assessment.
Paper 3	- Advanced Practical Skills. This paper requires candidates to carry out practical work in timed conditions. - This paper will consist of experiments drawn from different areas of the syllabus. - Candidates will answer all questions. Candidates will answer on the question paper. [100 marks]	Practical exam will cover 30% of the final marking of the assessment.

Assessment of Subject objectives (AO)

The assessment objectives listed below reflect those parts of the syllabus competences that will be assessed in the examination.

AO1 Knowledge with understanding

Candidates should be able to demonstrate knowledge and understanding of:

- Scientific phenomena, facts, laws, definitions, concepts and theories.
- Scientific vocabulary, terminology and conventions. (Including symbols, quantities and units)
- Scientific instruments and apparatus used in Physics, including techniques of operation and aspects of safety.
- Scientific quantities and their determination.
- Scientific and technological applications, with their social, economic and environmental implications.

The subject content defines the factual knowledge that candidates may be required to recall and explain.

Questions testing these assessment objectives will often begin with one of the following words: define, state, name, describe, explain (using your knowledge and understanding) or outline.

AO2 Handling information and solving problems

Candidates should be able to handle information and solve problems, using, written, symbolic, graphical and numerical forms of presentation to:

- Locate, select, organize and present information from a variety of sources.
- Translate information from one form to another.
- Manipulate numerical and other data.

- Use information to identify patterns, report trends and draw conclusions.
- Give reasoned explanations for phenomena, patterns and relationships.
- Make predictions and hypotheses.
- Apply knowledge including principles to new situations.
- Demonstrate an awareness of the limitations of physics theories and models.
- Solve problems.

These assessment objectives cannot be precisely specified in the syllabus content because questions testing such skills may be based on information which is unfamiliar to the candidate. In answering such questions, candidates are required to use principles and concepts that are within the syllabus and apply them in a logical, reasoned or deductive manner to a new situation.

Questions testing these assessment objectives will often begin with one of the following words: discuss, predict, suggest, calculate, and explain (give reasoned explanations and explain the processes of using information and solving problems) or determine.

AO3 Experimental skills and investigations

Candidates should be able to:

- Observe, give feedback, plan experiments and investigations.
- Collect, record and present observations, measurements and estimates.
- Analyse and interpret data to reach conclusions.
- Evaluate methods and quality of data and suggest possible improvements.
- Use ICT in solving problems.

3.5. Reporting to parents

Regular communication with parents is an essential component of the CBC. This is achieved through progress reports, parent-teacher meetings, and school community activities. Since the CBC emphasizes a wide range of learning outcomes: knowledge, skills, attitudes, and competences a single mark is insufficient to capture learner progress. Effective reporting should highlight areas where learners are excelling as well as those requiring improvement. Using descriptive feedback or simple rating scales (e.g., meeting expectations very well, meeting expectations, not meeting expectations) across knowledge, subject-specific skills, and competences provides a clearer and more comprehensive picture than numerical grades alone. This approach fosters meaningful parental involvement and supports learners' continued growth.

4. RESOURCES

The effective implementation of the Physics syllabus requires the provision and utilization of appropriate resources. These resources include both teaching and learning materials and qualified human personnel, without which the delivery of the syllabus would be ineffective.

Teachers are encouraged to utilize both locally available, laboratory and digital resources to create rich, relevant, and engaging learning experiences that promote creativity, problem-solving, and innovation.

4.1. Materials needed for implementation of this syllabus

Teaching and learning of Physics entails practical activities for better conceptualization of concepts and facts. The successful implementation of this curriculum requires Physics laboratories, textbooks, charts and ICT tools like computers and projectors.

However, there are some Physics concepts that cannot be easily explained and some experiments that cannot be done in our school laboratories due to safety reasons. The use of ICT in teaching and learning is vital. With ICT, these concepts can be visualised by use of videos, animations, simulations and virtual laboratories.

Similarly, both teachers and learners are encouraged to use the Internet for research as well as other ICT tools, for teaching and learning purposes.

4.2. Human resource

The effective implementation of this curriculum needs the joint collaboration of educators at all levels. Given the material requirements, teachers are expected to accomplish their noble role as stated above. School head teachers and directors of studies are required to follow-up and assess the teaching and learning of this subject due to their profile in the schools. These combined efforts will ensure bright future careers and lives for learners as well as the contemporary development of the country.

The following are some of the skills required for the teacher:

- Engage students in variety of learning activities.
- Use multiple teaching and assessment methods.
- Adjust instructions to the level of the learner.
- Creativity and innovation.
- Makes connections/relations with other subjects.
- Should have a high level of knowledge of the content.
- Effective discipline skills.
- Good classroom management skills.
- Good communicator.
- Guide and counselor
- Passion for learners in teaching and learning.

5. SYLLABUS UNITS

5.1. Presentation of the structure of the syllabus

Physics subject is taught and learned in upper secondary education as a core subject, i.e. in S4, S5 and S6 respectively. At every grade, the syllabus is structured in Topic Areas, Sub-topic Areas where applicable and then further broken down into Units. The Units have the following elements:

- Unit is aligned with the Number of sub-topics.
- Each unit has a key unit competence whose achievement is pursued by all teaching and learning activities undertaken by both the teacher and the learners.
- Each key unit Competence is broken into three types of learning objectives as follows:

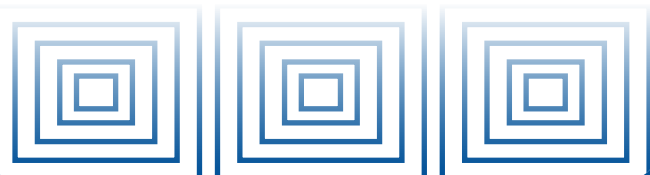
Type I: Learning Objectives relating to Knowledge and Understanding (Type I Learning Objectives are also known as Lower Order Thinking Skills or LOTS).

Type II and Type III: These Learning objectives relate to acquisition of Skills, Attitudes and Values (Type II and Type III Learning Objectives are also known as Higher Order Thinking Skills or HOTS).

- Each unit has content which indicates the scope of coverage of what a teacher should teach, and learner should learn, in line with stated learning objectives.
- Each unit suggests learning activities that are expected to engage learners in an interactive learning process as much as possible (Learner-centered and participatory approach). Note that the learning activities suggested are not exhaustive, the teacher may think of other possible learning activities for effective teaching and learning.

Finally, each unit is linked to other subjects, its Assessment Criteria and the Materials (or Resources) expected to be used in the teaching and learning process.

SENIOR FOUR



5.2. SENIOR FOUR

5.2.1. KEY COMPETENCES FOR SENIOR FOUR

- Apply the laws of reflection and refraction of light to describe image formation and solve related problems in daily life.
- Apply working principles of optical instruments to solve related problems.
- Apply principles of moments, equilibrium, and elasticity to solve real-life mechanical problems.
- Investigate the conservation laws of energy and momentum to solve collision-related problems.
- Analyze complex electric circuits using Kirchhoff's laws and appropriate measuring instruments.
- Analyze different sources and uses of energy in the world to promote sustainable, efficient, and environmentally responsible energy practices.
- Investigate the motion of bodies under projectile and circular motion using scientific reasoning.
- Investigate the interaction between electric charges and fields.
- Apply thermodynamic laws to explain energy transformations and their environmental effects.
- Analyze the structure and motion of celestial bodies to interpret astronomical phenomena and solar system.

5.2.2. SENIOR FOUR UNITS

TOPIC AREA: LIGHT			SUB-TOPIC: GEOMETRIC OPTICS	
S4 Physics	UNIT1: REFLECTION AND REFRACTION OF LIGHT		Nº of Periods: 28	
Key unit competence: Apply the laws of reflection and refraction of light to describe image formation and solve related problems in daily life.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Recognize apparent and real depths in refraction of light. • State the main characteristics and types of lenses. • Differentiate converging from diverging lenses. • Explain the refraction of light through lenses using ray diagrams. • Identify the nature and properties of an image formed by lenses at various object positions.	• Construct ray diagrams to represent images formed by spherical mirror and lenses. • Derive a spherical mirror formula. • Use graphical method to determine the focal length of a thin lens. • Derive the lens equation (formula) graphically. • Derive the formula for lateral displacement of light by a glass block.	• Value the role of reflection and refraction of light in everyday life. • Develop a positive attitude toward experimentation as a way of discovering physical laws through observation and evidence. • Exhibit honesty and accuracy in recording experimental data and drawing conclusions.	• Graphical derivation of mirrors formulae. • Refraction of light through a rectangular glass block. • Apparent and real depths. • Critical angle and total internal reflection. • Characteristics of lenses. • Types of lenses: converging and diverging lenses. • Refraction of light through lenses. • Ray drawing and properties of images formed by lenses for various object positions.	• In pairs, construct ray diagram showing the formation of images by a spherical mirrors for different object position. • Through guided inquiry, use the constructed ray diagrams to derive the spherical mirrors formula. • In pairs, construct ray diagrams for image formation by a thin lens.

• Describe the common defects of lenses and how they can be corrected. • Discuss the real-life applications of lenses. • Define the terms of magnification and power of a lens. • Explain a prism as it is used in optics. • Describe the refraction of light through a prism. • Explain the phenomenon of dispersion of light by a prism. • Discuss the real-life applications of refraction of light.	• Perform experiments related to reflection and refraction of light. • Use a graphical construction of an image to derive the formulae for either a focal length of a lens or of an effective focal length of a system of lenses. • Apply sign conventions while solving problems involving lenses. • Derive the formula for the angle of minimum deviation of a prism. • Solve problems involving reflection and refraction of light.	• Show respect for safety and materials by handling glass blocks, lenses, and other optical instruments responsibly.	• Graphical determination of focal length of lenses. • Thin lens equation, power of lens, magnification and sign convention. • Lens combination and effective focal length. • Derivation of lenses formulae. • Optical aberrations in lenses and mirrors. • Applications of combined lenses. • Refraction through prisms • Deviation of light rays by a glass prism. • Angle of minimum deviation and the determination of refractive index of a prism.	• Through guided inquiry and working in groups, develop a formula for a focal length of a thin lens and a system of two thin lenses in contact. • In pairs, analyze the properties of the constructed image from a spherical mirror. • In groups, perform different experiments involving reflection and refraction of light using locally available materials. • Using locally available materials and working in groups, investigate the difference between apparent and real depths.
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			• Dispersion of light by a prism. • Applications of total internal reflection of light (Optical fibers, Mirage formation, prism in optical instrument).	• In groups, solve problems involving reflection and refraction of light. • Projects: In pairs or groups, use the locally available materials to construct an optical fiber, a lens, and a periscope by either using prisms or mirrors. • Use simulations or video to demonstrate total internal reflection in optical fibers.
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Link to other subjects: Biology (Myopia, hypermetropia, and use of lenses), Mathematics (Functions and equations)

Assessment criteria: Learner properly conducts experiments and projects related to reflection and refraction of light, describes the properties of the image formed by spherical mirrors and lenses and correctly solves problems involving reflection and refraction of light in varied scenarios.

Teaching and learning materials: concave and convex lenses, candles or source of light, optical bench, screen or Cardboard, rectangular glass blocks, prism, water, drawings and optical pins, soft boards, sheets of white paper, mathematical set, scissors or razors, manilla paper and glue.

TOPIC AREA: LIGHT			SUB-TOPIC: GEOMETRIC OPTICS	
S4 PHYSICS	UNIT 2: OPTICAL INSTRUMENTS			Nº of periods: 23
Key unit competence: Apply working principles of optical instruments to solve related problems.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Define an optical instrument. • Describe image formation by a camera, simple and compound microscope and telescope. • Differentiate between a microscope and a telescope. • Define a magnifying power of an optical instrument.	• Use simple optical devices such as magnifying glasses, cameras, microscopes, and Telescope correctly and safely. • Compare image characteristics (size, orientation, and position) formed by different optical instruments.	• Value the real-life applications of simple and compound microscope and telescope. • Demonstrate perseverance and patience while aligning lenses or adjusting distances to obtain clear images.	• Meaning of an optical instrument. • Image formation by a camera. • Image formation by simple and compound microscope. • Magnifying power of microscope. • Astronomical telescope.	• Through think-pair-share, discuss the working principle of a lens camera, microscope and telescope. • Working in groups, use computer simulations to investigate how an image is formed by a lens camera. • Through group discussion with guided inquiry, construct a diagram for image formation in microscopes and telescope. • In pairs, discuss the working principle of a human eye as an optical instrument.

• Describe the working principle of microscopes and telescopes. • Discuss the real-life applications of optical instruments (microscope, telescope and human eye). • Discuss the defects of vision and ways of correcting them.	• Construct ray diagrams showing image formation in various optical instruments. • Investigate the causes of vision defects and demonstrate their correction using lenses.	• Demonstrating honesty and accuracy in observation exemplified by recording experimental data faithfully without falsifying results.	• Magnifying power of Telescope. • Human eye as single lens system. • Defects of vision and their correction.	• Working in pairs through guided inquiry, develop a formula for magnifying power of a microscope and telescope. • Through group discussion, by using microscope or telescope perform an experiment to investigate the properties of an image formed. • In pairs, discuss the eye defects and analyze their ways of correction . • Through group discussion, solve problems involving optical instruments. • Project: Working in pairs or group, design and construct a microscope and a telescope using locally available materials.
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Link to other subjects: *Biology (Human eye, microscope).Mathematics (Functions and equations)*

Assessment criteria: *Students appropriately performs experiments and projects involving optical instruments; accurately describes the working principles and real-life applications of optical instruments; perfectly constructs ray diagrams of image formation and investigates the properties of image from optical instruments and correctly solves problems involving optical instruments.*

Teaching and learning materials: *Converging lens, diverging lens, Microscope, Camera, telescopes, binoculars, Scissor, Corrugated paper, Strong glue, mathematical set and optical bench.*

TOPIC AREA: MECHANICS			SUB-TOPIC: STATICS	
S4 Physics	UNIT 3: MOMENTS AND EQUILIBRIUM OF BODIES			Nº of Periods: 22
Key unit competence: Apply principles of moments, equilibrium, and elasticity to solve real-life mechanical problems.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Define a moment of a force about a point. • State the principle of moments about a point. • Describe moments of a couple of forces and torques. • State a condition of equilibrium of a body about an axis. • Differentiate between stable, unstable and neutral equilibrium.	• Perform an experiment to verify the principle of moments. • Determine a center of mass of an object of regular shape. • Perform an experiment to determine Young's modulus of a given material.	• Demonstrate accuracy and patience when performing experiments on moments and equilibrium. • Value teamwork and cooperation when conducting group investigations or constructing experimental setups.	• Moment of a force about a point. • Principle of moments of force. • Moment of a couple of forces and torques. • Condition for equilibrium of a body about an axis. • Types of equilibrium: stable, unstable and neutral.	• Through think-pair-share, define a moment of force about a point. • In pairs, perform an experiment to verify the principle of moments. • Through group discussion, describe moment of a couple of forces and torques. • Through inquiry-based experiments, determine a mass of a meter ruler and weight of a given object.

• Define a center of gravity. • Discuss real life applications of moments. • Differentiate strain from stress. • Explain the meaning of a Young's modulus.	• Interpret a stress-strain graph of a given material. • Solve problems involving moments and equilibrium.	• Appreciate the relevance of moments and equilibrium in maintaining stability of structures and mechanical devices in daily life. • Show respect for evidence and logical reasoning when interpreting experimental results or verifying the principle of moments.	• Centre of gravity and the total weight. • Applications of moment (Ladder, see saw, beam). • Strain, stress and Young's moduli.	• Through group discussion, perform an experiment to determine the location of center of gravity of a given material with regular / irregular shape. • In pairs, perform experiments to demonstrate the difference between stable, unstable and neutral equilibrium. • Through group discussion, analyze the working principles of ladder, seesaw and beam as real-life applications of moments. • Through inquiry-based experiment, determine a Young's modulus of a given wood meter ruler.
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				• Through group discussion, solve problems involving moments of forces and torques, equilibrium of bodies, stress, strain and Young's modulus. • Project: Working in group, construct a seesaw and a beam balance.
Link to other subjects: Mathematics (torque calculations, geometry), and Biology (body joints and levers in movement).				
Assessment criteria: Learner effectively performs experiments and projects on moments of forces (torques) and Young's modulus; accurately explains the underlying principles and real-life applications; correctly interprets stress-strain graphs and solves problems related to moments, equilibrium of bodies, stress, strain, and Young's modulus.				
Teaching and learning materials: Meter rule, strong thread, spring balances, slotted weights, retort stand, Wooden knife edge, Wooden block, an irregular lamina, Plumb line and drawing pin or a nail.				

TOPIC AREA: MECHANICS			SUB-TOPIC: ENERGY AND MOMENTUM	
S4Physics	UNIT 4: CONSERVATION OF LINEAR MOMENTUM AND COLLISION		Nº of Periods: 25	
Key unit competence: Investigate the conservation laws of energy and momentum to solve collision-related problems.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Define work, energy and power. • Describe the mathematical expression (formula) for potential and kinetic energy and power. • State the work-energy theorem. • State the principle of conservation of mechanical energy.	• Interpret the conservation of mechanical energy through the exchange of kinetic energy into potential energy and vice versa. • Perform an experiment to verify the work-energy theorem. • Perform an experiment to verify the conservation of linear momentum.	• Demonstrate precision and accuracy in measurements and calculations during experiments. • Appreciate the importance of analyzing data to draw conclusions about momentum conservation. • Observe lab safety rules while performing collision experiments.	• Concepts of Work, Energy and Power. • Mathematical expression of potential energy (elastic gravitation), kinetic energy, Work energy theorem and power. • Conservation of mechanical energy. • Linear momentum.	• Through guided inquiry, derive the formula for collisions and work-energy theorem. • In pairs, interpret the exchange of kinetic to potential energy of an object thrown vertically upward and infer from it a conservation of mechanical energy. • Through group discussions, use computer simulations to illustrate the conservation of linear momentum. • In pairs, use computer simulations to verify the law of conservation of mechanical energy.

• Define a linear momentum. • Describe the principle of conservation of linear momentum. • Differentiate between elastic and inelastic collision.	• Use mathematical formula to describe elastic and inelastic collision. • Solve problems involving mechanical energy, linear momentum and collisions.	• Develop collaboration and teamwork while working with peers to design and carry out experiments.	• Conservation of linear momentum. • Collision (elastic and inelastic collision).	• Through think-pair-share, define and differentiate elastic from inelastic collisions. • Through group discussion, solve problems involving mechanical energy, linear momentum and collisions • Searching from internet, discuss the real-life applications of conservation of linear momentum and collision. • Project: Use a conservation principle of momentum and Newton's first law to simulate rocket propulsion.
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Link to other subjects: Mathematics (vector analysis, motion graphs), Chemistry (molecular collisions), ICT (simulation and data analysis), and Physical Education (momentum in sports)

Assessment criteria: Learner accurately performs experiments to verify conservation of mechanical energy, conservation of linear momentum, work-energy theorem and correctly interprets and solves problems involving mechanical energy, linear momentum and collisions.

Teaching and learning materials: Projectors, computers, simulations, animations / simulations, diagrams and charts

TOPICAREA:ELECTRICITY			SUB-TOPICAREA:CURRENT ELECTRICITY	
S4 Physics	UNIT 5: DIRECT CURRENT CIRCUIT			Nº. of periods:23
Key unit competence: Analyze complex electric circuits using Kirchhoff's laws and appropriate measuring instruments.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Define electric current, drift velocity, current density, voltage and electromotive force. • Explain the relationship between current, charge and time. • Describe internal resistance and how they affect terminal potential difference.	• Measure accurately current, voltage, and resistance using appropriate instruments. • Construct and test electrical circuits with resistors connected in series, parallel, and mixed configurations. • Apply Kirchhoff's laws to solve numerical problems and analyze complex DC circuits.	• Demonstrate safety awareness and responsibility when handling electrical components and instruments. • Collaborate effectively with peers when designing and analyzing circuits.	• Current, Drift Velocity, and Current Density. • Electromotive force (emf), Terminal potential and total internal resistance. • Sources of electric current and electric receptors/appliances. • Internal and external resistance, potential difference across a cell.	• Group discussion on the meaning of current, voltage, and resistance using concept maps. • Brainstorming on everyday devices that use electricity and classify them as sources or receptors. • Constructing circuits with resistors in series, parallel, and mixed arrangements using breadboards. • Measuring current and voltage using ammeters and voltmeters in different parts of the circuit.

• Distinguish between series, parallel, and mixed circuit connections. • State Kirchhoff's junction and loop laws. • Describe the purpose and principles of operation of ammeters, voltmeters, galvanometers, Wheatstone bridge and potentiometer.	• Determine unknown resistance using a Wheatstone bridge and potentiometer. • Explain the difference between potential difference and electromotive force. • Evaluate the advantages and disadvantages of series and parallel arrangements of batteries and resistors in practical circuit applications.	• Show accuracy, patience, and perseverance during practical investigations. • Appreciate the role of electricity in technological and societal development.	• Connection of electrical current source and resistors either in series or parallel or mix-up. • Kirchhoff's laws (loop rule and junction rule). • Measuring instruments: (Ammeter and Voltmeter, Galvanometer, Wheatstone Bridge, Potentiometer).	• Determining internal resistance of a cell experimentally. • Using a Wheatstone bridge to measure unknown resistance. • Using a potentiometer to compare emf of cells. • Solving numerical problems applying Ohm's law and Kirchhoff's laws. • Using circuit simulation software (e.g., PhET) to test circuit configurations before physical setup. • Students predict what will happen when additional components are added to a circuit (e.g., extra bulb or battery), then test and reflect.
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				Evaluating advantages and disadvantages of series vs. parallel configurations in real-world scenarios (e.g., home wiring, flashlights).
Link to other subjects: Chemistry (Electrochemical cells as current sources, electron flow, oxidation–reduction reactions inside batteries), Technical and Vocational Education (Electrical wiring, use of tools and safety procedures in real installations.), Mathematics (Use of algebra, ratios, proportionality (Ohm's law), simultaneous equations for Kirchhoff's laws, graph interpretation (V–I characteristics))				
Assessment criteria: Learner can easily apply Kirchhoff' laws to analyze and solve circuit problems; Correctly measures current, voltage, and resistance using appropriate instruments; Accurately constructs circuits with resistors in series, parallel, and mixed configurations.				
Materials: Ammeter, voltmeter, ohmmeter, Rheostat, conductors and resistors, batteries, connecting wires.				

TOPIC AREA: ENERGY			SUB-TOPIC AREA: ENERGY AND ITS APPLICATIONS	
S4 Physics	UNIT 6: SOURCES OF ENERGY IN THE WORLD		Nº. of periods:16	
Key Unit Competence: Analyze different sources and uses of energy in the world to promote sustainable, efficient, and environmentally responsible energy practices.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Explain the importance of energy in daily life and economic development. • Classify energy sources as renewable and non-renewable, giving relevant examples. • Describe the characteristics of different energy sources. • Explain how energy is converted from one form to another and how it is transmitted for use.	• Compare renewable and non-renewable energy sources based on efficiency, availability, and sustainability. • Analyze case studies or data on global and local energy consumption. • Construct charts, diagrams, or models showing energy conversions (e.g., solar to electrical energy).	• Appreciate the importance of clean and renewable energy sources. • Demonstrate responsibility in the use and conservation of energy resources. • Show interest and curiosity in exploring sustainable energy technologies.	• Meaning and importance of energy. • Classification of energy sources (renewable and non-renewable energy sources). • Characteristics of energy sources (availability, cost, efficiency, and environmental effects).	• Demonstrate energy conversion using simple models (e.g., solar panel lighting a bulb). • Debate: “Fossil fuels should be replaced by renewable energy in Rwanda.” • Individually, reflect on how you can conserve energy at home, school, and in the community.

• Discuss the environmental and economic impacts of different energy sources. • Describe ways of conserving energy and promoting sustainable use of energy resources. • Identify and explain the future trends and technologies in energy production and utilization.	• Investigate methods of energy conservation through practical or simulated activities. • Use ICT tools to research and present information on modern energy technologies. • Evaluate the environmental effects of different energy sources and suggest possible solutions.	• Develop a positive attitude toward environmental protection and sustainable development. • Value collaboration and creativity in solving real-life energy challenges.	• Conversion and transmission of energy. • Environmental and economic impacts of energy use. • Energy conservation and sustainable use. • Future trends in energy.	• Group activity (Role Play): Students act as energy experts advising the Rwanda Energy Group (REG) on the best type of energy source or project to invest in. • Project: Reducing the Harmful Effects of Non-Renewable Energy.
Link to other subjects: Geography (Mapping potential renewable energy areas in Rwanda), Chemistry (combustion of fuels, electrochemical cells, and energy changes in chemical reactions).				
Assessment criteria: Learners are assessed based on their understanding of types, characteristics, and uses of energy sources, their ability to analyze environmental and economic impacts, and their practical application of energy conservation principles.				
Materials: Charts and posters showing types of energy sources, Maps of energy resources in Rwanda and the world, CDs, light bulb, connecting wires, and voltmeter, small solar panel.				

TOPIC AREA: MECHANICS			SUB-TOPIC AREA: MOTION IN TWO DIMENSIONS	
S4 PHYSICS	UNIT 7: PROJECTILE AND UNIFORM CIRCULAR MOTION		N ^o . of periods: 27	
Key unit competence: Investigate the motion of bodies under projectile and circular motion using scientific reasoning.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Identify the main characteristics of projectile motion and related terms. • Describe different types of projectile motion. • Interpret graphs of projectile motion. • State expressions for maximum height, time of flight, and horizontal range. • Define circular motion.	• Illustrate examples of projectile motion. • Resolve projectile motion in horizontally and vertically components. • Derive equations of projectile motion • Determine the maximum height and horizontal range in projectile motion.	• Appreciate the relevance of projectile and circular motion in technology and sports. • Show concern of the effects of gravity on body projected at an angle to the horizontal. • Demonstrate accuracy and teamwork during problem-solving and experiments.	• Meaning of projectile motion and related terms. • Types of projectile (vertical, horizontal and oblique). • Graphs of projectile motion. • Expressions of projectile motion (horizontal range and maximum height, time of flight).	• Using demonstration/ Experiment, Launch small balls at different angles and measure the horizontal range and maximum height. Record and compare results. • Use a spinning object (e.g., a rubber stopper on a string or conical pendulum) to measure centripetal force and velocity.

• Explain the relationship between linear and angular quantities. • Identify the forces acting on a body in circular motion. • Outline real-life applications of projectile and circular motion.	• Derive equations of circular motion. • Apply concepts of projectile, circular motion to different real life. • Solve problems involving projectile and circular motion.		• Applications of projectile motion. • Introduction to Circular Motion (Definition and types). • Quantities in Circular Motion (angular displacement, linear and angular velocity, period and frequency, angular and linear acceleration). • Equations of Circular Motion with constant acceleration. • Relationship Between Linear and Angular Quantities.	• Individual/Pair, Plot displacement-time and velocity-time graphs from experimental or simulated data. • Individual, Solve numerical problems involving time of flight, maximum height, and range for different types of projectiles. • Solve problems relating linear and angular velocity, centripetal and tangential acceleration.
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			• Forces in Uniformly Accelerated Circular Motion (Tangential, Centripetal and resultant forces). • Application of circular motion (vertical and horizontal, conical pendulum, spinning dryer, road banking).	• Working in groups, identify examples of projectile motion in sports (e.g., football, basketball) and discuss the factors affecting the motion.
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Links to other subjects: Physical sports (trajectory of basketball, football, netball, golf, darts), Mathematics (Uses algebra, trigonometry, and geometry), Geography (Earth's rotation and orbital motion in physical geography).

Assessment criteria: Learners will be assessed on their ability to explain and apply concepts of projectile and circular motion, solve numerical problems accurately, interpret motion graphs, and conduct experiments or simulations safely.

Materials: ball, Conical pendulum (bob, thread and fixed point), road banking, whirling water in bucket, Computer or projector, Small balls, marbles, Meter rulers, measuring tapes, or meter sticks, Stopwatches.

TOPIC AREA: ELECTROSTATICS			SUB-TOPIC AREA: INTERACTIONS OF ELECTRIC CHARGES	
S4 PHYSICS	UNIT 8: ELECTRIC FORCE AND ELECTRIC FIELD		N ^o . of periods: 25	
Key unit competence: Investigate the interaction between electric charges and fields.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Describe the concept of electric charge. • State Coulomb’s law. • Explain the principle of superposition of electric forces. • Describe electric field lines. • Explain electric potential and potential energy.	• Calculate the electric force between point charges using Coulomb’s law. • Determine the resultant electric field due to multiple charges. • Sketch electric field patterns for various charge distributions.	• Appreciate the role of electrostatic principles in daily life and technology. • Demonstrate accuracy and responsibility in performing electrostatics experiments.	• Electric charge and Coulomb’s law. • Principle of superposition of electric forces. • Electric field and electric field lines. • Electric field due to a single electric charge. • Electric field pattern due to more than one charge (resultant electric field). • Electric Potential.	• In whole-class discussion, brainstorm common electrostatic phenomena such as attraction between rubbed balloons or lightning. • In pairs or groups, perform experiments to illustrate Coulomb’s law. • Working individually or in pairs, solve numerical and graphical problems by apply Coulomb’s law and the principle of superposition of forces. • Use simulations (e.g., PhET) or field-mapping experiments to observe and draw field patterns around charges.

• Define a capacitor. • Explain charging and discharging processes of capacitors. • Identify the importance of capacitors.	• Solve problems related to electric potential and potential energy. • Compute equivalent capacitance in series and parallel combinations.	• Show curiosity and teamwork while investigating electric fields and capacitor applications.	• Lightning and lightning arrestors. • Capacitor (definition, combinations, energy stored, charging and discharging). • Applications of capacitors.	• In pairs or small groups, connect capacitors in series and parallel, measure the equivalent capacitance, and demonstrate charging and discharging using a voltmeter and power supply. • Working individually, calculates equivalent capacitance, stored charge, and energy in capacitors through problem-solving exercises. • In group research and presentations, explore and present real-life applications of capacitors.
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Links to other subjects: Chemistry (electrolysis, dye cells), Geography (formation of clouds and lightning), Mathematics (Vector addition).

Assessment criteria: Learners correctly explain electric charge and apply Coulomb's law, demonstrate the superposition of forces, illustrate electric field lines and describe fields due to single or multiple charges, analyze the motion of charges in uniform electric fields, and explain the functioning of capacitors in series and parallel.

Materials: plastic rods, glass rod, electroscope, metal plate, balloons, Capacitors, Electroscopes, Ammeter, Voltmeter, Multimeter, and battery.

TOPIC AREA:HEAT AND THERMODYNAMICS			SUB-TOPIC AREA:THERMAL EFFECTS	
S4 Physics	UNIT 9:LAWS OF THERMODYNAMICS			No.of periods:26
Key unit Competence: Apply thermodynamic laws to explain energy transformations and their environmental effects.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Differentiate internal energy and total energy of a system. • Explain the work done by expanding gas. • State the first law of thermodynamics. • State the second law of thermodynamics. • Explain thermodynamic processes in heat engines.	• Calculate work, heat, and energy changes in thermodynamic systems. • Interpret P–V diagrams for various processes. • Evaluate the performance and efficiency of engines and refrigerators. • Apply thermodynamic principles to propose energy-saving and sustainable solutions.	• Appreciate applications of heat engines in human development. • Acquire capacity to work with heat engines and reduce their effect on climate change. • Show responsibility toward mitigating climate change and environmental degradation.	• Thermodynamic systems. • Work done by an expanding gas. • Paths between thermodynamic states. • Internal energy and thermal energy. • First law of thermodynamics. • Thermodynamic processes: Isothermal, Isochoric and isobaric processes.	• Working groups to differentiate first and second laws of thermodynamics. • Work in groups to investigate changes in energy and work-done for a thermodynamic process and present findings. • Work in groups to evaluate change in energy and work done during the Carnot cycle and report.

Describe the relationship between thermodynamics and climate change and the greenhouse effect.			- Adiabatic processes for ideal gas. - Second law of thermodynamics. - Applications of second law of thermodynamics: Carnot engine, diesel engine and refrigerator. - Efficiency of heat engine. - Climate change and greenhouse effect.	- Draw and interpret P–V diagrams for isothermal, isobaric, isochoric, and adiabatic processes. - Project: Designing and testing a simple solar cooker. - Solve problems related to efficiency of refrigerators and heat engines. - Perform an experiment to investigate work done by an expanding gas using a syringe and air.
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Links to other subjects: Chemistry (heat of reactions, energy changes), Geography (climate change, energy balance)

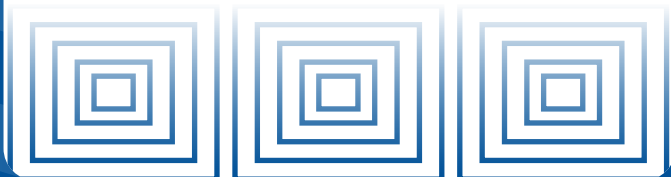
Assessment criteria: Learner explains the concepts of work, internal energy, and the laws of thermodynamics, distinguishes between isothermal, isobaric, isochoric, and adiabatic processes using appropriate diagrams and examples, applies the first and second laws of thermodynamics to solve numerical and real-life problems, and analyzes the efficiency and operation of heat engines and refrigerators, interprets the relationship between thermodynamic principles, the greenhouse effect, and climate change, while demonstrating a positive attitude toward energy conservation and environmental sustainability.

Materials: Syringes, Thermometers, pressure gauges, Models or diagrams of heat engines, refrigerators, and the Carnot cycle, videos or simulations of thermodynamic processes, Charts or posters showing the greenhouse effect.

TOPIC AREA:ASTROPHYSICS			SUB-TOPIC AREA:EARTH AND SPACE	
S4 Physics.	UNIT 10:GENERAL STRUCTURE OF THE SOLAR SYSTEM		Nº. of periods: 16	
Key unit competence: Analyse the structure and motion of celestial bodies to interpret astronomical phenomena and solar system.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Explain scales for estimate astronomical distances. • Explain the phenomenon of eclipse and phases of the moon. • Differentiate inner, outer planets, comets, meteorites and asteroids. • Explain stars patterns. • Describe celestial coordinates used to locate celestial objects.	• Illustrate the phenomenon of eclipse by considering the relative positions of the sun-moon-earth system • Outline the positions of the eight planets with the sun. • Apply celestial coordinates. • Use ICT tools and astronomical software to simulate or observe celestial phenomena.	• Appreciate the importance of orbital motion of the earth to human life and activities. • Acquire the ability to observe the universe and identify planets and stars. • Collaborate effectively during group discussions and observational activities.	• Astronomical scales • Sun-Earth-Moon system: (eclipses, and phases of the moon) • The Solar system: (Inner planets, outer Planets, comets, meteorites, asteroids). • Star patterns: constellations. • Celestial coordinates: Horizontal system (hour angle, zenith angle); Equatorial system (right ascension, declination).	• Work in groups, discuss the meaning of astronomical scales and compare units such as the light-year, astronomical unit, and parsec. • Using demonstration, Use a light source (Sun), a globe (Earth), and a ball (Moon) to demonstrate lunar phases and eclipses. • Use instrument telescope and Galileo telescope to observe planets and present the findings.constellation.

				• Search for simulations from the Internet on planetary motion. • Use stellarium software simulate position of planets, stars and constellation.
Links to other subjects: Geography (climate change and seasons),telecommunication(radio,Global positioning system (GPS),)				
Assessment criteria: Learners demonstrate the ability to analyze celestial bodies and their interactions by describing astronomical scales, the Sun–Earth–Moon system, and solar system components. They accurately interpret star patterns and use celestial coordinates to locate objects.				
Materials: Telescopes and globe, light source, ball, Stellarium software.				

SENIOR FIVE



5.3. SENIOR FIVE

5.3.1. KEY COMPETENCES FOR SENIOR FIVE

- Apply the principles of simple harmonic motion to explain, model, and solve real-life situations involving periodic motions.
- Analyse different effects of damped and forced oscillations on systems.
- Apply principles of mechanical wave propagation to solve problems and interpret real-life phenomena.
- Analyse real-life situations involving fluid motion using principles of fluid mechanics.
- Apply the laws of gravity to analyse planetary and satellite motion.
- Interpret atomic models, atomic structure and the process of electron emission in practical contexts.
- Analyse the dual nature of light to explain its wave and particle properties and their applications.
- Evaluate the application of analog and digital signal transmission to analyse and propose solutions that enhance efficiency and reliability in modern communication systems.
- Investigate how magnetic fields are generated and used in practical devices such as DC motors and mass spectrometers.
- Apply the concepts and postulates of special relativity in solving related problems.
- Investigate the interference and diffraction of light using practical experiments to explain its wave properties.

5.3.2. SENIOR FIVE UNITS

TOPIC AREA: OSCILLATIONS AND WAVES			SUB-TOPIC: OSCILLATIONS	
S 5 PHYSICS	UNIT 1: SIMPLE HARMONIC MOTION (SHM)		Nº of Periods: 25	
Key unit competence: Apply the principles of simple harmonic motion to explain, model, and solve real-life situations involving periodic motion in physical systems.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Describe a simple harmonic motion. • Explain the parameters (frequency, angular frequency, period, and amplitude) of SHM. • Give examples of simple harmonic oscillators.	• Represent a SHM graphically. • Illustrate SHM as the projection of uniform circular motion. • Derive equations of SHM executed by a simple pendulum, spring-mass system, and liquid in a U-shaped glass tube. • Interpret graphs of energies in SHM.	• Demonstrate patience, perseverance, and precision while taking measurements of an experiment. • Develop logical reasoning while analysing observed data to relate different parameters of SHM.	• Simple harmonic motion (definition, characteristics, examples). • Kinematics of simple harmonic motion (equation of motion, displacement, velocity, acceleration).	• In pairs, use AI-based simulations to illustrate SHM as the projection of uniform circular motion. • Through guided inquiry, derive the equations of SHM for various simple harmonic oscillators. • Through guided inquiry, derive a formula for kinetic, potential, and total mechanical energies for SHM. • In group discussions, explain changes in energy and energy conservation for a SHM and represent it graphically.

• Describe the kinematics (by equations) of different simple harmonic oscillators. • Give the formula for potential, kinetic, and total energy relating to a SHM. • Explain the changes in energy and energy conservation in SHM. • Explain the superposition of harmonic motions of the same frequency.	• Derive equations for the superposition of SHMs with the same frequency. • Represent graphically the superposition of SHMs with the same frequency. • Perform experiments relating to SHM. • Solve problems involving SHM.	• Value the real-life applications of SHM. • Develop teamwork and collaboration to work effectively with peers in performing and interpreting experiments.	• Energy in simple harmonic motion (potential, kinetic, and total energy). • Energy changes and conservation in oscillating systems. • Simple harmonic oscillators (simple pendulum, elastic pendulum, liquid oscillator). • Superposition of harmonic motion with the same frequency.	• Through inquiry-based experiments, use the concepts of SHM to determine the gravitational acceleration, force constant of a spring, effective mass of the spring, and the moment of inertia of a meter ruler about its centre. • In pairs, use AI-based simulations to verify the conservation of mechanical energy of SHM. • Through guided inquiry, derive the formula for a resultant motion from the superposition of two SHM with the same frequency and graph its resultant motion. • Through group discussion, solve SHM problems.
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Links to other subjects: *Mathematics (Trigonometric and differential functions). Biology and health sciences (vibrations in ear structures for hearing). Geography and environmental sciences (seismic waves during earthquakes). Chemistry (vibrations of molecules).*

Assessment criteria: *Learner accurately describes SHM, effectively performs related experiments, and correctly interprets graphs and solves problems involving SHM.*

Materials: *Two-meter rules, one retort stand with a clamp, two pieces of wood that can fit between the clamps, thread, a stopwatch, one pendulum bob, a set of slotted weights, springs, a horizontal table, and AI-based computer simulations.*

TOPIC AREA: PERIODIC MOTION			SUB-TOPIC AREA: OSCILLATIONS	
S 5 physics	UNIT 2: DAMPED AND FORCED OSCILLATIONS			No. of periods: 20
Key unit Competence: Analyse different effects of damped and forced oscillations on systems.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Define a damped oscillation. • Describe types of damped oscillations. • Explain examples of damped oscillators. • Describe Energy Transfer in a damped system.	• Perform experiments to demonstrate the damping of oscillating systems. • Analyse damped oscillators. • Design a control mechanism to mitigate the resonance. • Derive an equation for the energy of a damped oscillation.	• Appreciate applications of damped and forced oscillators in life. • Acquire scientific attitudes for interpreting the resonance. • Develop a commitment to precision in measurement and calculation.	• Damped oscillations. • Equation of damped oscillation and its solution. • Types of damped oscillations (underdamped, critically damped and over damped oscillation). • Energy change in damped oscillations • Forced oscillation. • Equation of forced oscillation.	• In groups, perform an experiment to demonstrate damping of oscillating systems. • Through guided inquiry, use AI based simulations and describe SHM executed by simple pendulum, spring-mass system and liquid in a U-shape tube and derive corresponding equations.

• Explain a forced oscillation and its energy exchanges. • Describe resonance and give examples.	• Analyse a forced oscillation using energy exchange. • Analyse graphically forced oscillations. • Represent graphically underdamped, critically damped, and overdamped oscillations. • Solve problems involving damped and forced oscillations.	• Acquire knowledge for analysing and modelling physical processes. • Develop an appreciation for experimental evidence and rigorous analysis of system parameters.	• Energy change in forced oscillations. • Effect of resonance on a system. • Examples of resonance.	• Working in groups, describe a forced oscillation and represent it graphically. • Through guided inquiry, use AI-based computer simulations to perform an experiment on resonance, suggest more examples of it and evaluate its uses and effects. • Project: Design and development of a simple system based on the knowledge from damped and forced oscillations.
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Links to other subjects: Mathematics (differential equations, graphs and functions), Chemistry (Molecular vibrations), Biology (Vibration and damping in biological systems such as the human ear (sound reception) and heartbeat regulation), Geography (Explains seismic waves and earthquake oscillations).

Assessment criteria: Learners will be assessed on their ability to explain and differentiate types of damped and forced oscillations, derive and interpret their equations, and describe the effects and examples of resonance in real-life situations, solve numerical problems related to oscillatory motion and interpret corresponding graphs.

Materials: Simple pendulum (string and bob), masses, springs, Retort stand, clamp, and stopwatch, masses, AI based computer simulations

TOPIC AREA: OSCILLATIONS AND WAVES			SUB-TOPIC: WAVES	
S 5 PHYSICS	UNIT 3: PROPAGATION OF MECHANICAL WAVES		Nº of Periods: 26	
Key unit Competence: Apply principles of mechanical wave propagation to solve problems and interpret real-life phenomena.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Define a mechanical wave • Explain the characteristics of a wave. • Distinguish between transverse and longitudinal waves with examples. • Describe progressive (travelling) and stationary (standing) waves. • Give examples of progressive and standing waves in real-life situations.	• Derive a relationship between wavelength, frequency, Period and velocity. • Derive the equations of travelling and stationary waves. • Interpret the equations of progressive and stationary waves.	• Appreciate the role of waves in real-life applications. • Develop a scientific attitude of observation, inquiry, and logical reasoning. • Demonstrate teamwork, respect and responsibility during experimental activities.	• Wave concept. • Types of waves (Transverse and longitudinal). • Characteristics of a wave (amplitude, frequency, period, wavelength, crests and troughs). • Progressive wave. • Equation of a progressive/ travelling wave. • Stationary waves.	• Through group discussion, use the ripple tank or AI-based computer simulations to demonstrate the wavefronts, crests and troughs of a wave. • In pairs, use either springs or AI-based computer simulations or animations to illustrate and differentiate transverse and longitudinal waves. • Through guided inquiry, derive the equations of travelling and stationary waves.

• Describe a standing wave. • Explain the formation of stationary waves on a vibrating string.	• Represent waves graphically. • Interpret wave parameters from graphs. • Perform experiments to demonstrate transverse and longitudinal waves. • Solve problems involving wave equations and relationships among wave quantities.	• Show curiosity and interest in exploring wave phenomena in nature and technology.	• Equation of stationary/ standing waves. • Wave on a vibrating string. • Relationship between frequency, tension, and mass per unit length characterising standing waves on a vibrating string. • Real-life applications of travelling and standing waves.	• Working in pairs, use either AI-based computer simulations or a string to illustrate the process of the formation of stationary waves and explain concepts of nodes and antinodes. • Through group discussion, use either videos or laboratory apparatus to perform Melde's experiment. • Through group discussion with the use of any AI tool, develop a descriptive report on real-life applications of travelling and standing waves. • Through group discussion, solve problems involving wave propagation.
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Links to other subjects: Mathematics (Trigonometric and differential functions). Biology and health sciences (sound transmission in the ear, vibrations in the vocal cords). Geography and environmental sciences (seismic waves in earthquakes, tsunamis and tidal waves). Chemistry (vibrational motion of atoms and molecules).

Assessment criteria: Learner accurately describes travelling and standing waves and correctly derives and applies equations and formulas to solve problems involving mechanical wave propagation.

Materials: One 120 cm cotton rope, an electric vibrator, an alternating current power supply, a hanging mass of 20 g to set a tension in the rope, a fixed pulley, a switch, ripple tank/ shallow water tray, water, Vibrator/ ripple tank motor, 2 connecting wires, source of voltage/ power supply, ripple tank lamp, 2 elastic bands, white screen, video for Melde's experiment.

TOPIC AREA: MECHANICS			SUB-TOPIC: FLUID MECHANICS	
S 5 PHYSICS	UNIT 4: FLUID MECHANICS		Nº of Periods: 20	
Key unit competence: Analyse <i>real-life situations involving fluid motion using principles of fluid mechanics.</i>				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Explain the variation of pressure in liquids and its measurement using a U-tube manometer. • Describe surface tension, cohesive and adhesive forces, and their effects on liquid surfaces and capillarity. • Differentiate types of fluid flow by relating them to Reynolds' number and flow conditions.	• Derive the formula for pressure at a given depth, surface tension and capillary rise. • Demonstrate the difference between laminar and turbulent flow experimentally. • Derive continuity and Bernoulli's equations.	• Show curiosity and interest in exploring the behaviour of liquids in everyday phenomena. • Appreciate the importance of fluid mechanics in real-life experiences.	• Variation of pressure in a liquid. • U-tube manometers • Surface tension. • Cohesive and adhesive forces and the shape of liquid surfaces. • Capillarity. • Introduction to fluid motion. • Types of fluid flow (laminar, turbulent).	• Through guided inquiry, derive and explain the formula for pressure inside a liquid at a certain depth. • Through group discussion, perform an experiment to demonstrate the difference between laminar and turbulent flow and give conditions for these flows using Reynolds' number. • Through think-pair-share, differentiate cohesive from adhesive forces and explain their effects on the shape of liquid surfaces.

• Explain fundamental equations of fluid motion, including continuity, Bernoulli's principle and terminal velocity. • Illustrate practical applications of fluid motion in everyday and technological situations.	• Derive the formula for terminal velocity. • Derive the formula for a venturi meter. • Apply fundamental equations to solve problems involving fluid mechanics.	• Develop a positive attitude toward applying physics principles to explain natural occurrences. • Work collaboratively and responsibly during group activities.	• Continuity equation (conservation of mass in fluid flow). • Bernoulli's principle. • Venturi meter. • Motion of objects through a viscous fluid. • Applications of fluid motion.	• Through guided inquiry, derive and interpret the formula for surface tension, capillarity rise, Bernoulli's principle, continuity, terminal velocity and venturi meter. Use any AI-based simulation to illustrate the application of Bernoulli's equation. • Through group discussion, solve fluid mechanics problems. • Through research-based groups with the help of any AI tool, discuss and report on real-life applications of fluid mechanics. • Project: Design and development of a simple system based on fluid mechanics.
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Links to other subjects: Mathematics (Equations and functions). Biology and health sciences (Blood Circulation and Plant Transport). Geography and environmental sciences (Understanding laminar and turbulent flow in rivers, sediment transport, and erosion). Chemistry (Use of pipettes, burettes, and flow through capillary tubes).

Assessment criteria: Learner correctly derives and applies equations for fluid mechanics to solve related problems and accurately interprets real-life applications of fluid mechanics.

Materials: U-tube manometer, transparent containers, water, capillary tubes, spheres of different sizes, viscous liquids (oil, glycerin), stopwatch, ruler, spray bottles, Bunsen burner and carburetor demo.

TOPIC AREA: GRAVITATION AND SPACE PHYSICS			SUB-TOPIC AREA: PLANETARY AND SATELLITE MOTION	
S 5 Physics	UNIT 5: MOTION IN ORBITS		N ^o . of periods: 20	
Key Topic Competence: Apply gravitational laws to analyse planetary and satellite motions.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• State Newton's law of universal gravitation. • Describe a gravitational field and its properties around Earth and other celestial bodies. • Explain gravitational potential energy, kinetic energy, and total mechanical energy in orbital motion. • State Kepler's three laws of planetary motion.	• Apply Newton's law of gravitation to calculate the gravitational force between two masses. • Determine gravitational field strength and gravitational acceleration in different regions relative to Earth's surface. • Analyse the energy changes involved in orbital motion.	• Demonstrate curiosity and interest in exploring space science and celestial motion. • Show appreciation of scientific contributions in understanding the universe. • Exhibit teamwork and respectful communication during group problem-solving and investigations.	• Newton's law of gravitation. • Gravitational field. • Variation of gravity above and below the earth surface. • Work done in planetary motion. • Energy in orbital motion (Gravitational potential energy, Kinetic energy and total energy).	• Using AI-based simulations and animations, discuss in groups observable examples of motion in space (e.g., the Moon orbiting Earth, satellites, planets around the Sun). • Working in groups, use either a video, AI based computer simulation or animations to illustrate and analyze planetary motion, gravity, and satellite launches. • Through group discussion, solve problems involving gravitational force, acceleration due to gravity, and orbital velocity.

- Describe how Kepler's third law can be verified using orbital data. - Explain first, second, and third cosmic velocities. - Describe the types of satellites and their main applications.	- Use Kepler's laws to solve problems related to planetary and satellite orbits. - Evaluate satellite types and orbits in relation to their applications and societal needs. - Differentiate between first, second, and third cosmic velocities through numerical and conceptual tasks. - Verify Kepler's third law using orbital data.	- Recognise the importance of satellite technology in improving daily lives. - Acknowledge the need for responsible and sustainable development of space technologies to protect the environment and society.	- Kepler's laws of planetary motion. - Verification of Kepler's third law of planetary motion. - Cosmic velocity (first, second and third). - Satellites and their applications.	- Working in groups with the help of any AI-tool, discuss and report on the applications of satellites in communication, weather prediction, and navigation. - In group discussions, compare and contrast rockets and satellites on their working principles, structures, and functions. - In pairs, use the orbital data to verify Kepler's third law of planetary motion. - Through group discussion, analyse the energy change in orbital motion.
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Links to other subjects: Geography (understanding the solar system, satellite applications in GPS, weather monitoring, mapping and remote sensing), Chemistry (electrons' motion).

Assessment criteria: Learner correctly states and explains Newton's law of universal gravitation, Kepler's laws of planetary motion and applies them to interpret and solve problems involving orbits of planets or satellites.

Materials: AI based Simulations and animations (e.g., PhET Interactive Simulations: My Solar System, Gravity and Orbits), Videos and documentaries on space exploration, planetary motion, and satellite launches

TOPIC AREA: QUANTUM PHYSICS			SUB-TOPIC AREA: ATOMIC PHYSICS	
S 5 Physics	UNIT 6: ATOMIC MODELS			N ^o . of periods: 21
Key unit Competence: Interpret atomic models, atomic structure and the process of electron emission in practical contexts.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Describe the structure of the atom and its main subatomic particles. • Explain J.J. Thomson's, Rutherford's, and Bohr's atomic models. • Explain the concept of energy levels and the formation of spectral lines. • Describe the process of thermionic emission.	• Compare different atomic models based on their assumptions, strengths, and limitations. • Represent the arrangement of subatomic particles within various atomic models. • Interpret atomic spectra to relate energy levels with electron transitions.	• Recognise the importance of atomic theory in technological and scientific advancements. • Demonstrate curiosity and interest in exploring atomic and subatomic phenomena.	• J.J. Thomson's Plum Pudding model. • Rutherford's atomic model. • Bohr's atomic model. • Energy levels and spectral lines. • Thermionic emission. (thermo-electronic emission).	• Using a guided discussion, the teacher leads a discussion on the discovery of subatomic particles (electrons, protons, neutrons). • In groups, discuss Thomson, Rutherford and Bohr atom models (enumerate similarities and differences).



• Explain the applications of cathode rays in devices such as oscilloscopes and television tubes. • Describe the significance and applications of atomic models in modern science and technology.	• Demonstrate thermionic emission using a simple experimental setup or simulation. • Analyse the operation of cathode ray tubes and their applications in oscilloscopes and televisions.	• Collaborate effectively with peers during group discussions and practical investigations. • Appreciate the contribution of scientists such as Thomson, Rutherford, and Bohr to the understanding of atomic structure.	• Applications of cathode rays (oscilloscope and TV tubes). • Applications of atomic models.	• Calculate energy differences between electron orbits from given data on spectral lines. • Use AI-based computer simulations to demonstrate thermionic emission using a filament circuit. • Working in groups, analyse energy level diagrams to relate electron transitions with emission of spectral lines.
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Links to other subjects: Chemistry (Understanding atomic and molecular energy levels, electron transitions, atomic structure and atomic spectra)

Assessment criteria: Learner clearly describes the atomic models, the structure of an atom and correctly solves problems involving energy levels and spectral lines.

Materials: Cathode ray tube (CRT) apparatus, AI based computer simulations and videos.

TOPIC AREA: QUANTUM THEORY			SUB-TOPIC AREA: MODERN PHYSICS	
S 5 Physics	UNIT 7: WAVE AND PARTICLE NATURE OF LIGHT		Nº of periods: 25	
Key Unit Competence: Analyse the dual nature of light to explain its wave and particle properties and their applications.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Explain the concept of black-body radiation. • Describe Planck’s quantum theory and its role in the development of modern physics. • Explain the photon theory of light and the idea of wave-particle duality. • State expressions for the energy, mass, and momentum of a photon.	• Demonstrate the photoelectric effect using appropriate AI- based computer simulations or laboratory setups. • Calculate the energy, frequency, and wavelength of photons using Planck’s equation. • Analyze graphs of stopping potential versus frequency to interpret photoelectric results.	• Demonstrate curiosity and open-mindedness toward new scientific ideas and discoveries. • Appreciate the importance of quantum theory in explaining phenomena beyond classical physics. • Acknowledge the contributions of scientists such as Planck, Einstein, and Compton to modern physics.	• Black-body radiation. • Planck’s quantum theory. • Photon theory of light. • Wave-particle duality. • Energy and momentum of photon. • Photoelectric effect. • Photon, work function and Plank’s constant.	• In pairs, brainstorm on the limitations of classical physics in explaining black-body radiation and the photoelectric effect. • Group discussion on the concepts of photons, wave-particle duality, and their implications in modern physics. • Demonstration or AI-based simulation of the photoelectric effect using a virtual lab or experimental setup.

• Explain the photoelectric effect and its experimental observations. • Explain Einstein's photoelectric equation and how it relates photon energy to electron emission. • Describe the Compton effect and its significance in demonstrating photon interactions. • Explain the wave nature of matter and its experimental evidence. • Describe the working principle and application of the electron microscope.	• Compare experimental observations with predictions from wave and particle theories of light. • Apply Einstein's photoelectric equation to solve related problems. • Interpret the Compton effect and electron diffraction results to support the concept of wave-particle duality. • Use an electron microscope or AI-based simulation to explore the wave nature of matter.	• Value the role of quantum theory in technological advancements. • Respect evidence-based reasoning as the foundation of scientific understanding.	• Einstein's equation of photoelectric effect. • Factors affecting photoelectric emission. • Application of the photoelectric effect. • Compton Effect. • Photon interactions. • The wave nature of matter. • Electron microscope.	• Problem-solving exercises involving calculations of photon energy, wavelength, frequency, and kinetic energy of emitted electrons. • Video analysis or AI-based simulation showing Compton scattering and electron diffraction experiments. • Hands-on activity or virtual exploration of the operation and applications of an electron microscope.
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Links to other subjects: Chemistry (atomic spectra, electron transitions, and the quantisation of energy levels).

Assessment criteria: Learner correctly applies principles of quantum and wave theories to explain fundamental concepts of black-body radiation, Planck's quantum theory, photon theory, and wave-particle duality, and calculate observable quantities.

Materials: Simulations demonstrating black-body radiation, photon emission, and diffraction patterns (e.g., PhET simulations), Photoelectric effect apparatus (light source, photocell, voltmeter, and ammeter), Spectrometer and prism.

TOPIC AREA: DIGITAL TECHNOLOGY		SUB-TOPIC AREA: ANALOG AND DIGITAL SIGNALS		
S 5 physics	UNIT 8: SIGNALS AND COMMUNICATION SYSTEMS			N° of periods: 14
Key unit competence: Evaluate the application of analog and digital signal transmission to analyse and propose solutions that enhance efficiency and reliability in modern communication systems.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Distinguish between amplitude, frequency, and phase modulation. • Compare the advantages and disadvantages of analog and digital systems.	• Illustrate the basic components and flow of information in a communication system using diagrams. • Demonstrate how signal conversion occurs using (ADC) and (DAC) simulations. • Compare the characteristics of AM, FM, and PM through practical examples or virtual experiments.	• Recognise the importance of digital technology in improving communication efficiency. • Value digital communication innovations and their contribution to science, education, and industry.	• Introduction to communication systems. • Analog and digital signals. • Conversion of signals. • Modulation and transmission (Amplitude Modulation (AM), Frequency Modulation (FM), and Phase Modulation (PM)).	• Work in groups and discuss analog and digital systems of communication. • Role-play the advantages of the digital system and compare with the analog system. • Use an oscilloscope or AI-based computer simulation to visualise analogue and digital waveforms. • Discuss the difference between telephone and radio systems.

• Trace the evolution of communication technologies from PTT to modern digital systems. • Identify the main parts of a simple telecommunication block diagram and their functions.	• Analyse the transmission and reception process in radio and mobile communication systems. • Justify the use of digital over analog systems under certain conditions. • Construct and interpret a simple communication system diagram.	• Demonstrate a sense of responsibility and safety when working with communication devices and electrical equipment.	• Mobile phone communication systems. • Radio communication systems. • Evolution of communication technologies (Post, Telegraph, and Telephone (PTT) systems, Transition to modern mobile and digital communication.	• Role play in groups about types of modulation. • Demonstration/ Simulation to show ADC and DAC conversion processes. • Roles play the communication of the microphone, antenna, and present the report. • Draw and label parts (transmitter, channel, and receiver); explain the function of each using real-life examples.
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Links to other subjects: *Geography* (Study of satellite communication, GPS technology), *ICT* (how data and voice are transmitted through computer networks, mobile phones, and the Internet).

Assessment criteria: *Learner correctly identifies and describes the components of a communication system, differentiates between analog and digital signals, performs or simulates modulation and signal conversion, and explains the evolution and applications of communication technologies, interprets a block diagram of a telecommunication system, appreciates the importance of communication in modern life, and shows responsible attitudes toward the use of communication devices.*

Materials: *Charts or posters showing types of communication systems, radio, mobile phone, television, Simulation software, oscilloscope and signal generator (if available). Examples of analog and digital communication devices (old radio vs smartphone), Samples or photos of telecommunication towers and satellites, Reference textbooks and internet resources.*

TOPIC AREA: ELECTROMAGNETISM			SUB-TOPIC AREA: MAGNETISM	
S 5 Physics	UNIT 9: MAGNETIC FIELD AND MAGNETIC FORCES			No. of periods: 23
Key unit Competence: Investigate how magnetic fields are generated and used in practical devices such as DC motors and mass spectrometers.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Identify sources of magnetic fields and explain how they are generated. • Describe the properties of magnetic fields, flux, and forces on charges and conductors. • Explain the principles behind the motion of charged particles in magnetic fields and their applications.	• Calculate forces on moving charges and current-carrying conductors. • Analyse experimental setups such as Thomson's e/m experiment and velocity selectors.	• Appreciate the applications of magnetism in technology. • Develop curiosity to explore electromagnetic devices. • Practice safety when working with magnets and current-carrying conductors.	• Magnetic fields and their sources (magnetic field, properties of magnetic field lines, and sources of magnetic fields). • Current-carrying conductors as sources of magnetic fields (a straight current-carrying wire, a circular loop of wire, a solenoid). • Force on a moving charge in a magnetic field. • Magnetic flux and Gauss's law for magnetism.	• By demonstration, show magnetic fields using permanent magnets, solenoids, and wires to visualize field lines and direction. • Perform experiments such as Thomson's e/m experiment simulation. • Calculate forces on moving charges and current-carrying conductors using numerical problems and diagrams.

• Define a magnetic flux. • State Gauss's law for magnetism. • Illustrate the force on a current-carrying conductor. • Describe force and torque on a current loop in a magnetic field. • Define the magnetic moment of a coil or current loop.	• Demonstrate understanding of DC motor operation through diagrams or simulations. • Solve problems involving magnetic fields and magnetic forces.		• Force on a current-carrying conductor. • Motion of charged particles in magnetic fields. • Applications of motion of charged particles (Velocity selector, Thomson's e/m experiment, mass spectrometers). • Force and torque on a current loop in a magnetic field. • Magnetic moment of a coil or current loop. • Direct-current motor.	• In a group or pair with the help of any AI-tool, explore and explain real-life applications of magnetism in devices like DC motors, velocity selectors, and mass spectrometers. • Individual/Pair work, use computer-based simulations to observe the motion of charged particles in magnetic fields and the functioning of electromagnetic devices. • Project: Design and development of a simple system based on magnetic field and magnetic forces.
Links to other subjects: <i>Chemistry (electrochemistry and atomic structure). Geography and Environmental Sciences (Earth's magnetic field and navigation).</i>				
Assessment criteria: <i>Learner correctly analyses and explains magnetic phenomena by identifying sources and directions of magnetic fields, calculating forces on moving charges and current-carrying conductors, describing the motion of charged particles in magnetic fields and interpreting applications like velocity selectors, mass spectrometers, and DC motors.</i>				
Materials: <i>Permanent magnets, solenoids, wires, ammeters, power supplies, Iron filings, compasses, AI-based computer simulations.</i>				

TOPIC AREA: RELATIVITY AND PARTICLE PHYSICS			SUB-TOPIC: CONCEPTS AND POSTULATES OF SPECIAL RELATIVITY	
S 5 PHYSICS	UNIT 10: INTRODUCTION TO SPECIAL THEORY OF RELATIVITY		Nº of Periods: 17	
Key unit competence: Apply concepts and postulates of special relativity in solving related problems.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Explain relativity and its significance in Physics • Describe the concepts of space, time, mass, and their interdependence in motion • Define a frame of reference • Differentiate between inertial and non-inertial frames of reference. • State the Galilean transformation equations.	• Analyze scenarios involving simultaneity and reference frames to predict observations in different frames. • Derive equations for time dilation, length contraction, energy-mass equivalence, relativistic mass and relativistic relation between energy and momentum.	• Show interest and willingness to learn and understand surprising ideas such as time dilation and length contraction. • Demonstrate critical thinking and reasoning when evaluating physics phenomena from different reference frames.	• Meaning of relativity. • Concept of space, time and mass. • Concept of frame of reference • Galilean equation of transformation. • Postulates of special theory of relativity.	• Through group discussion, describe the concepts of space, time, mass, and their interdependence in motion. • Through think-pair-share and using AI-based animations, define a frame of reference and differentiate inertial from non-inertial frames of reference. • Through guided inquiry, derive equations for time dilation, length contraction, energy-mass equivalence, relativistic mass and relativistic relation between energy and momentum.



• State the postulates of the special theory of relativity. • Explain the consequences of special relativity, including time dilation, length contraction, mass-energy equivalence, and the concept of simultaneity.	• Derive Galilean equations of transformation. • Solve problems involving concepts and principles of special relativity.	• Develop teamwork and collaboration with peers during group activities.	• Consequences of special relativity (time dilation, length contraction, mass and energy equivalence and relativistic mass). • Concept of simultaneity.	• Working in pairs and using AI-based simulations, illustrate the time dilation. • Working in groups and using computer-based simulations, discuss the concept of simultaneity. • Through group discussion with the help of any AI-tool, state and interpret the postulates of special relativity. • Through group discussion, solve problems involving concepts of special relativity.
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Links to other subjects: Mathematics (Equations and functions).

Assessment criteria: Learner accurately describes the concepts of special relativity, including time dilation, length contraction, energy-mass equivalence, relativistic mass and relativistic relation between energy and momentum; effectively derives Galilean equations of transformation and correctly solves problems involving special relativity.

Materials: AI-based animations and simulations, a computer and a projector.

TOPIC AREA: ELECTROMAGNETIC WAVES			SUB-TOPIC AREA: LIGHT WAVES	
S 5 PHYSICS	UNIT 11: INTERFERENCE AND DIFFRACTION OF LIGHT		Nº of Periods: 20	
Key unit competence: Investigate the interference and diffraction of light using practical experiments to explain its wave properties.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Explain evidence showing that light behaves as a wave. • Describe the nature and properties of electromagnetic waves. • Differentiate between mechanical and electromagnetic waves. • Explain the characteristics of light waves such as wavelength, frequency, amplitude, and velocity.	• Perform experiments to demonstrate the interference and diffraction of light. • Calculate wavelength and fringe spacing from experimental data. • Use drawings to interpret interference and diffraction patterns.	• Appreciate the importance of light waves in life. • Acquire scientific techniques, reasoning and attitudes in analysing electromagnetic waves. Light as electromagnetic wave. • Conditions for interference of light. • Constructive and destructive interference. • Double-slit experiment. • Prediction of the location of interference fringes.	• Light as electromagnetic wave. • Conditions for interference of light. • Constructive and destructive interference. • Double-slit experiment. • Prediction of the location of interference fringes.	• Discuss in groups the conditions necessary for interference. • Show a video or an AI-based simulation to introduce the idea of light interference and diffraction. • Devise and perform a double-slit experiment and estimate the wavelength of light.

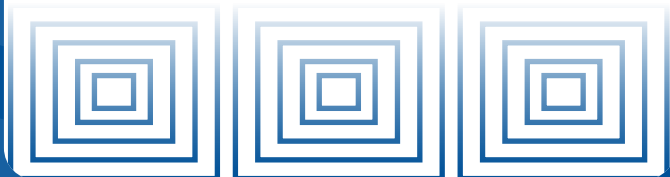
• Explain the principle of superposition of waves. • State the conditions necessary for interference to occur. • Describe interference and diffraction of light, including the double-slit experiment and the resulting fringe patterns.	• Use the double-slit setup to verify the wave nature of light. • Analyse data by relating intensity distribution to fringe spacing and coherence. • Solve problems involving interference and diffraction of light.	• Acquire scientific reasoning and attitudes in interpreting the concepts of interference. • Show curiosity and teamwork in performing and analysing light experiments.	• Intensity distribution of fringe pattern. • Diffraction of light. • Diffraction pattern. • Diffraction grating. • Application of interference and diffraction of light.	• Solve problems relating to the double slit experiment. • Through demonstration, pass monochromatic light through a single slit and measure the diffraction pattern on a screen. • Conduct a research on Any AI-tool light interference and diffraction.
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Links to other subjects: Chemistry (absorption/emission spectra and colour changes in chemical solutions), Mathematics (trigonometry, algebra, and graphing).

Assessment criteria: Learner accurately performs and critically interprets experiments on interference and diffraction, clearly describes the wave nature of light, effectively applies the superposition principle, precisely calculates wavelength and fringe spacing, and coherently communicates findings

Materials: Diffracting or Slits, slit holder, screen, source of light, white screen, Laser pointer or monochromatic light source, retort stand, Screen or white wall, ruler or measuring tape, AI-based computer simulations.

SENIOR SIX



5.4. SENIOR SIX

5.4.1. SENIOR SIX KEY COMPETENCES

- Demonstrate the behavior and properties of sound waves in various physical situations.
- Evaluate the effects of torque, moment of inertia, and angular momentum in rotational systems.
- Demonstrate the production of induced electromotive force and its applications in daily life.
- Analyze the characteristics and behavior of alternating current in electrical devices and power systems.
- Apply the principles of atomic nuclei and radioactivity in energy production and safe radiation handling.
- Analyse the nature of particle and their interactions.
- Investigate how X-rays and LASERs are produced and used safely in different fields.
- Demonstrate understanding and safe application of radiation principles in medical imaging and therapy.
- Investigate how different electronic materials and devices work illustrating their structure and role in circuits.
- Analyze the structure and evolution of the Universe demonstrating the nature and properties of galaxies and stars.

5.4.2. SENIOR SIX UNITS

TOPIC AREA: OSCILLATIONS AND WAVES			SUB-TOPIC AREA: WAVES.	
S6 Physics		UNIT 1: SOUND WAVES		No. of periods:26
Key unit Competence: Demonstrate the behavior and properties of sound waves in various physical situations.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Define key terms related to sound waves. • Describe the characteristics and properties of sound waves. • Explain how sound is produced and transmitted through different media.	• Measure the frequency, wavelength, and speed of sound in different media. • Demonstrate the reflection, refraction, and interference of sound waves through experiments. • Determine the relationship between frequency and length in vibrating strings and pipes.	• Appreciate the importance of sound in communication, music, and technology. • Demonstrate curiosity and creativity in exploring sound phenomena. • Collaborate effectively during experiments and group investigations on sound.	• Production of sound waves. • Characteristics of sound waves (amplitude, loudness, frequency, pitch, quality and overtones, frequency limits of audibility). • Speeds of sound in various medium. • Properties of sound waves (reflection, refraction, diffraction and interference).	• Discuss common sources of sound and create a concept map showing how sound is produced and transmitted. • Use AI virtual labs or simulations to visualize wave reflection, interference, and diffraction. • In groups, conduct experiments using stretched strings or open/closed pipes to study frequency–length relationships and harmonics.

• Identify factors affecting the speed and intensity of sound. • Interpret resonance, harmonics, and the doppler effect in sound waves.	• Calculate sound intensity, resonance frequencies, and doppler shifts. • Construct graphs showing the behaviour of sound waves. • Apply principles of sound in solving practical and technological problems.	• Exhibit responsibility in handling sound-producing instruments and laboratory equipment. • Value the role of physics in improving sound-related technologies and daily life.	• Vibrations in strings and pipes (frequency and length pipe and harmonics in strings and pipes). • Resonance . • Sound intensity. • Doppler effect in sound waves. • Applications of sound waves.	• In groups, generate beats using two tuning forks of slightly different frequencies and observe the effect. • Working in groups, solve numerical problems on resonance, Doppler effect, and sound intensity. • Use appropriate AI tools to explore properties of sound waves. • Project: Investigating Resonance and Frequency in Air Columns.
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Links to other subjects: *Biology (Study of the human ear, hearing process, and effects of sound intensity on health), Music (sound production, resonance, harmonics, and pitch) Geography (sound waves in echo sounding and seismic studies).*

Assessment criteria: *Learner describes correctly how sound is produced, transmitted, and detected in different media, perform an experiment to illustrate propagation of sound wave and solve problems related to sound propagation in different media.*

Materials: *Tuning forks of different frequencies, Resonance tube apparatus, Stopwatches or timers, String and sonometer apparatus, Signal generator and oscilloscope, AI-Simulation software (e.g., PhET "Sound"), Multimedia clips showing interference, beats, and Doppler effect.*

TOPIC AREA: MECHANICS			SUB-TOPIC AREA: DYNAMICS	
S6 Physics	UNIT 2: ROTATIONAL MOTION			Nº of periods: 24
Key unit Competence: Evaluate the effects of torque, moment of inertia, and angular momentum in rotational systems.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Define key terms related to rotational motion. • Describe the kinematics of rotational motion. • Describe the kinetics of rotational motion. • Explain the concept of the moment of inertia in rotational systems. • Relate work, power, and energy to rotational motion.	• Measure angular displacement, velocity, and acceleration using experimental setups in rotational motion. • Determine the moment of inertia of simple rotating bodies through calculation or experiment. • Calculate torque, angular momentum, and rotational kinetic energy from given data.	• Appreciate the importance of rotational motion in understanding mechanical systems and natural phenomena. • Demonstrate curiosity and persistence in exploring rotational dynamics through experiments.	• Rotational motion and its dynamics (Meaning and key terms). • Torque and rotational equilibrium. • Moment of inertia. • Angular momentum and its conservation.	• Discuss in groups examples of rotational motion in daily life (wheels, fans, planets, etc). • Demonstrate uniform and non-uniform rotational motion using a turntable, bicycle wheel, or fan. • Working in groups, solve numerical problems involving torque, angular momentum, rotational kinetic energy, and work-energy principles.

• Discuss the principles of angular momentum and rotational equilibrium. • Identify real-life applications of rotational motion in machines and natural phenomena.	• Demonstrate the relationship between torque and angular acceleration experimentally. • Solve problems related to rotational dynamics and equilibrium.	• Exhibit accuracy, patience, and responsibility when using laboratory equipment.	• Work, power, and energy in rotational motion (Work done by a constant torque, Rotational power, Rotational kinetic energy). • Kinetic energy of a rolling object. • Work–Kinetic energy theorem for rotational motion. • Applications of rotational motion.	• Use AI simulations or animations (e.g., PhET “Rotational Motion”) to visualize the relationships among angular variables and moments of inertia. • Determine the moment of inertia of various bodies (e.g., disk, ring, rod) experimentally and compare with theoretical values.
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Links to other subjects: Chemistry (Centrifugal separation and molecular rotations relate to rotational motion principles). Geography (Rotation of the Earth and its effects), Physical Education (Analysis of body rotation in sports like gymnastics, discus, and javelin).

Assessment criteria: Learners will be assessed based on their ability to explain and apply the concepts of rotational motion, distinguish between linear and rotational motion, solve related numerical problems, demonstrate practical skills in determining moment of inertia, work, power, and energy in rotational systems, and show understanding of angular momentum, equilibrium, and stability through experiments.

Materials: Physics simulation software (PhET), projector, computer, disk, ring, rod, turntable, bicycle wheel, fan.

TOPIC AREA: ELECTROMAGNETISM		SUB-TOPIC AREA: MAGNETIC EFFECTS OF ELECTRIC CURRENT		
S6 Physics	UNIT 3: ELECTROMAGNETIC INDUCTION			No. of periods: 26
Key unit Competence: Demonstrate the production of induced electromotive force and its applications in daily life.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Define key terms related to electromagnetic induction, such as magnetic flux, flux linkage, and induced emf. • Explain Faraday's and Lenz's laws of electromagnetic induction. • Describe how motion or a changing magnetic field produces an electromotive force (emf).	• Demonstrate experimentally the generation of an electromotive force (emf). • Measure magnetic flux, induced emf, and current in simple experimental setups. • Construct and test basic circuits illustrating self-induction and mutual induction.	• Appreciate the role of electromagnetic induction in modern electrical and energy technologies. • Demonstrate curiosity and creativity in exploring magnetic and electrical interactions through experiments.	• Faraday's experiments. • Faraday's law of electromagnetic induction (magnetic flux and magnetic flux linkage). • Faraday's law of electromagnetic induction (Statement and mathematical expression). • Motional electromotive force.	• Perform Faraday's experiment using a magnet and coil to observe induced current. • Demonstrate Lenz's law with a copper tube and magnet. • Illustrate mutual induction using two coils and a galvanometer. • In groups, derive mathematical expressions for Faraday's and Lenz's laws.

• Interpret the relationship between magnetic flux, rate of change of flux, and induced emf mathematically. • Distinguish between self-induction and mutual induction. • Discuss the operation and applications of devices based on electromagnetic induction.	• Calculate induced emf, inductance, and energy stored in inductors using appropriate formulas. • Analyse the performance of electromagnetic devices such as transformers and generators. • Investigate experimentally how design factors affect induced emf.	• Exhibit accuracy, patience, and responsibility when handling electrical instruments and conducting experiments. • Value the contribution of electromagnetic principles to sustainable energy solutions and technological innovation.	• Lenz's law. • Inductors and Inductance (Magnetic Flux, Induced EMF and self-mutual inductance, inductance of coil and energy). • Applications of electromagnetic induction (transformer, generators, eddy current).	• Individually or in pairs, solve numerical problems involving induced emf, magnetic flux linkage, and inductance. • Analyse energy transfer and efficiency in transformers. • Use simulations (PhET, Falstad, or Multisim) to visualize electromagnetic induction processes and flux changes. • Project: Building a Simple Transformer.
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Links to other subjects: Chemistry (Electrochemical processes and energy conversions in generators and batteries), Mathematics (Application of algebra, calculus, and trigonometric functions), ICT(Use of simulation or videos of electromagnetic induction).

Assessment criteria: Learners will be assessed based on their ability to explain and apply Faraday's and Lenz's laws of electromagnetic induction to different physical situations, demonstrate understanding of magnetic flux, flux linkage, and induced electromotive force (emf) through problem-solving and practical experiments, describe and analyze self and mutual inductance, and to illustrate how transformers, generators, and eddy currents operate based on electromagnetic induction principles.

Materials: Bar magnets, solenoids, iron nails, compass, connecting wires, galvanometer, ammeter, voltmeter, Dry cells or low-voltage AC power supply, copper wire, resistors, switches, PhET simulation or videos on electromagnetic induction, projector or computer for visual demonstrations.

TOPIC AREA: ELECTROMAGNETISM			SUB-TOPIC AREA: ELECTROMAGNETIC INDUCTION	
S6 Physics	UNIT 4: ALTERNATING CURRENT			Nº. of periods: 26
Key unit Competence: Analyze the characteristics and behavior of alternating current in electrical devices and power systems.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
- Define key terms related to alternating current. - Explain the generation and characteristics of sinusoidal alternating currents and voltages. - Describe the phasor representation of alternating quantities. - Distinguish between peak, average, and root mean square (r.m.s.) values of AC. - Interpret the behaviour of simple AC circuits containing resistors, inductors, and capacitors.	- Demonstrate the generation of alternating current using simple laboratory setups or simulations. - Measure peak, average, and r.m.s. values of AC voltage and current using appropriate instruments. - Construct simple AC circuits (R, L, C, RL, RC, and RLC) and observe phase relationships.	- Appreciate the importance of alternating current in modern technology and daily life applications. - Demonstrate curiosity and initiative in exploring the behaviour of AC circuits through experiments.	- AC sources and sinusoids. - Phasor representation. - AC. values, Root mean square values and peak values of A.C. - Simple circuits (R circuit, L circuit and C circuit). - Series circuits (RL circuit, RC circuit, LC circuit, RLC circuit).	- Use an oscilloscope or simulation to display sinusoidal waveforms and discuss amplitude, period, and frequency. - Discuss in groups the differences between direct current (DC) and alternating current (AC) using simple demonstrations. - Illustrate phasor diagrams and show the phase relationship between voltage and current in different circuits (R, L, C). - Derive expressions for instantaneous voltage and current in AC circuits.

- Describe the impedance, phase relationships, and resonance in series AC circuits (RL, RC, LC, RLC). - Discuss the power relationships in AC circuits, including power factor and energy transfer. - Illustrate the practical applications of alternating current in power transmission, electric motors, and communication systems.	- Calculate impedance, phase angle, and power factor in various AC circuit configurations. - Operate measurements from AC instruments such as oscilloscopes and multimeters. - Investigate experimentally the condition for resonance in an RLC circuit. - Analyse data from experiments to verify theoretical relationships between voltage, current, and phase in AC circuits.	- Exhibit responsibility, accuracy, and safety awareness when handling electrical apparatus. - Collaborate effectively with peers during experimental work and problem- solving discussions. - Value the contribution of AC principles to energy distribution, transportation, and communication systems.	- Power in an AC circuit. - Electrical resonance (series and parallel resonance). - Applications of AC (power transmission, electric motors, radio and communication).	- Solve problems on RMS values, impedance, phase difference, and power factor. - Use vector and phasor analysis to interpret circuit behaviour. - Use simulation software (PhET, Falstad, Multisim) to visualize current and voltage variations in AC circuits. - Watch videos or animations on the functioning of AC generators, transformers, and motors.
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Links to other subjects: Chemistry (application of electrochemistry concepts in AC-based electrolysis systems), Mathematics (application of trigonometric functions in sinusoidal wave forms).

Assessment criteria: Learner should be explain fundamental AC concepts, correctly describe and interpret phasor relationships and circuit behaviors in RLC circuit and apply AC circuit laws to calculate impedance, phase angle, power, and resonance frequency.

Materials: Function generator or AC power supply, Oscilloscope, multimeters, resistors, capacitors, and inductors of various values, breadboards and connecting wires, transformer kits, lamps, switches.

TOPIC AREA: ATOMIC PHYSICS			SUB-TOPIC: NUCLEAR PHYSICS	
S6 PHYSICS	UNIT 5: ATOMIC NUCLEI AND RADIOACTIVE DECAY		No of Periods: 25	
Key unit competence: Apply the principles of atomic nuclei and radioactivity in energy production and safe radiation handling.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Define key terms related to atomic nuclei. • Describe the nature of radioactivity and factors affecting nuclear stability. • Apply Einstein’s mass–energy relation to nuclear processes. • Distinguish between nuclear fission and fusion, and explain the principles governing each.	• Measure radiation levels using appropriate detectors. • Record data on radioactive decay to determine half-life experimentally. • Calculate mass defect, binding energy, and energy released in fission or fusion reactions. • Differentiate between types of radiation based on penetration power, ionization, and charge.	• Appreciate the importance of nuclear physics in energy production, medicine, and industry. • Demonstrate responsibility and caution when handling or discussing radioactive materials. • Show respect for environmental and human safety in the use of nuclear technology.	• Atomic nuclei-nuclide. • Radioactivity and nuclei stability. • Unified atomic mass. • Einstein’s mass-energy relation. • Equivalent of unified atomic mass in electro volt. • Mass defect, Binding energy and stability. • Nuclei fusion and fission.	• In groups, use the periodic table and isotope charts to find nuclear symbols, atomic numbers, and mass numbers, and to tell the difference between isotopes, isobars, and isotones. • Use computer or AI simulations to observe how unstable nuclei change during radioactive decay and to tell natural radioactivity from artificial radioactivity.

• Describe the processes of controlled and uncontrolled nuclear reactions and their implications. • Explain radioactive decay, half-life, and decay series. • Describe types of radiation (alpha, beta, gamma) • Discuss applications of radioactivity. • Explain safety precautions in handling radioactive materials.	• Demonstrate safe laboratory handling of radioactive sources. • Construct and balance nuclear equations for various decay processes. • Evaluate the risks and benefits of nuclear technologies based on scientific data.	• Promote awareness of the benefits and risks associated with radioactivity. • Display ethical consideration in the application of nuclear knowledge for peaceful purposes. • Develop curiosity and critical thinking about advancements in nuclear science and technology.	• Controlled and uncontrolled nuclear fusion. • Controlled fission (power production) and uncontrolled fission (nuclear weapons). • Radioactivity decay. • Radiation detectors. • Types and properties of emitted radiations. • Application of radioactivity. • Hazards and safety precautions of when handling radiations.	• Work in pairs to calculate average atomic mass, change atomic mass units into kilograms or electron volts, and link mass with energy using Einstein's formula. • Compare the binding energy of different nuclei using charts to find out which nuclei are most stable. • Use animations or AI simulations to show nuclear fusion and fission, and explain how energy is released and controlled in reactors and the Sun.
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Links to other subjects: <i>Mathematics (Exponential functions, graphical representation and quantitative analysis). Chemistry (Radioactive elements and isotopes). Biology and Health Sciences (Radiation effects on living cells). History (Use of nuclear fission in WWII, carbon dating).</i>				
Assessment criteria: <i>Learner correctly identifies and explains concepts pertaining to nuclear physics; accurately solves related problems; effectively interprets data and graphs, properly describes ways of detecting radiations based on their properties and critically evaluate their real-life applications together with associated risks and ethical implications.</i>				
Materials: <i>Periodic tables, isotope charts, Animations or simulations, Videos or documentary clips, Charts and diagrams for Binding energy, decay chains, and radiation types.</i>				

TOPIC AREA: MODERN PHYSICS			SUB-TOPIC AREA: PARTICLE PHYSICS	
S6 Physics	UNIT 6: NATURE OF PARTICLES AND THEIR INTERACTIONS		N ^o . of periods: 25	
Key unit Competence: Analyse the nature of particle and their interactions.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Explain the structure of matter. • Identify quarks and leptons as the fundamental building blocks of matter. • Classify elementary particles. • Explain the concept of antiparticles and antimatter and their applications.	• Use models or simulations to illustrate particle structure and interactions. • Apply conservation laws to analyze particle reactions and decays. • Interpret Feynman diagrams. • Solve simple problems involving mass–energy equivalence.	• Appreciate the importance of particle physics in explaining natural phenomena. • Show curiosity, teamwork, and critical thinking during research and presentation. • Exhibit awareness of the ethical and environmental implications of particle physics applications.	• Structure of matter (constituents of matter, atomic number, mass number, and isotopes, subatomic particles, quarks and leptons as fundamental building blocks). • Classification of elementary particles (leptons, hadrons, quarks). • Antiparticles and antimatter. • Fundamental interactions (forces).	• Guided discussions on the structure of matter using atomic and subatomic models. • Analyze charts showing classification of particles (leptons, hadrons, quarks). • Group work to compare the four fundamental interactions using a table or chart. • Class discussion on conservation laws illustrated with worked examples.

• Describe the four fundamental interactions. • Interpret simple Feynman diagrams representing particle interactions. • Explain particle reactions, decays, and the Standard Model, highlighting their applications in real life.	• Conduct research or presentations on particle accelerators, detectors, and modern discoveries.		• Laws of conservations (baryon number, lepton number, strangeness, charge, energy, linear momentum). • Particle reactions and decays (pair production and annihilation, beta decay, electron capture, Feynman diagrams, mass-energy equivalence). • Particle accelerators and detectors. • The standard model of particle physics. • Applications of elementary particle.	• Use PhET simulations to observe pair production or beta decay. • In groups, draw and interpret simple Feynman diagrams for particle interactions. • Research and presentations on particle accelerators, detectors, or CERN discoveries.
Links to other subjects: Chemistry (Atomic structure, isotopes, and radioactivity), Mathematics (Application of algebra and energy-mass equations), Geography (Environmental impact of nuclear energy and radiation on ecosystems).				
Assessment criteria: Learners clearly explain the structure of matter and identify its fundamental particles, classify and compare different types of elementary particles and interactions, accurately apply conservation laws in analyzing particle reactions and decays, demonstrate practical skills such as interpreting Feynman diagrams, using simulations, and presenting research findings clearly.				
Materials: PhET simulations or animations of particle decays, Charts or posters showing conservation laws, Internet resources or textbooks on particle physics, educational videos on accelerators.				

TOPIC AREA: MODERN PHYSICS			SUB-TOPIC AREA: ELECTROMAGNETIC WAVES	
S6 Physics	UNIT 7: X-RAYS AND LASER RADIATIONS			N ^o . of periods: 21
Key unit Competence: Investigate how X-rays and LASERs are produced and used safely in different fields.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Describe the nature of X-rays and LASERs, and locate them in the electromagnetic spectrum. • Explain the production processes and characteristic spectra of X-rays and LASERs. • Describe the properties of X-rays and LASER light, emphasizing their physical characteristics.	• Analyze diagrams or simulations showing the production of X-rays and LASER beams. • Interpret and differentiate spectra of X-rays and visible LASER light. • Investigate and present findings on the uses of X-rays and LASERs in medicine, industry, and communication.	• Appreciate the importance of X-rays and LASERs in advancing science, medicine, and technology. • Show curiosity and teamwork when exploring applications and safety of electromagnetic radiation.	• Introduction to X-Rays and LASERs (Definition and nature of X-rays and LASER). • X-rays and LASERs in the electromagnetic spectrum. • Production of X-rays. • Properties of x-rays. • X-rays spectrum: origins and characteristic features. • Applications of X-rays (in medicine, industry, Security and scientific research).	• Teacher-led discussion on electromagnetic spectrum showing where X-rays and LASERs fall. • Use of diagrams or models to explain the production of X-rays in an X-ray tube and of LASERs using stimulated emission. • Group comparison of properties and spectra of X-rays and LASERs. • Class brainstorming on different applications in medicine, industry, and research.

• Discuss the applications, dangers, and safety precautions related to X-rays and LASERs, and compare the two forms of radiation.	• Demonstrate awareness of safe handling and protective measures through classroom simulations or projects.	• Demonstrate responsibility and caution in activities involving high-energy radiation or light sources. • Develop ethical awareness about the safe and appropriate use of X-ray and LASER technologies in society.	• Dangers of X-rays and Safety Precautions. • Production of a LASER beam. • Properties of LASER light (monochromatic, coherent, directional). • Applications of LASERs (In Medicine, Industry, Communication and research). • Dangers and Safety Precautions of LASERs. • Comparison of X-Rays and LASERs.	• Learners observe or use PhET simulations or videos showing X-ray production and LASER emission. • Draw labeled diagrams of X-ray tubes and LASER setups. • Research group activity: "Uses of LASERs and X-rays in Rwanda's hospitals or industries." • Role-play on safety precautions for radiographers or engineers working with these devices. • Debate on the benefits versus dangers of using X-rays and LASERs.
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Links to other subjects: *Chemistry (X-ray diffraction used to study crystal structures), Biology (Medical imaging and radiation therapy applications), Geography (Use of remote sensing and LASER-based mapping), ICT (Use of LASERs in optical communication and data transmission).*

Assessment criteria: *Learners clearly define and explain the nature, production, and properties of X-rays and LASERs, distinguish their similarities and differences, correctly describe and analyze spectra, identify practical applications, and apply appropriate safety precautions.*

Materials: *Charts showing the electromagnetic spectrum, Diagrams or models of X-ray tubes and LASER devices, Multimedia or PhET simulations on X-ray and LASER production, Textbooks, posters, and internet resources.*

TOPIC AREA: ELECTROMAGNETIC WAVES			SUB-TOPIC: RADIATION	
S6 PHYSICS	UNIT 8: RADIATION AND MEDICAL IMAGING		Nº of Periods: 23	
Key unit competence: Demonstrate understanding and safe application of radiation principles in medical imaging and therapy.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Describe the nature and types of radiation, including ionizing and non-ionizing radiation. • Explain the interaction of radiation with matter. • Define physical, biological, and effective half-life. • Explain radiation dosimetry. • Describe the principles of sound relevant to ultrasound imaging.	• Solve radiation dosimetry problems. • Select appropriate imaging techniques based on clinical needs. • Analyze risks and benefits of radiation use in diagnosis and therapy.	• Appreciate the importance of medical imaging in patient care. • Develop a safety-conscious mindset toward radiation exposure. • Show responsibility and ethical awareness when using medical imaging techniques.	• Nature and types of radiation: Ionizing and non-ionizing. • Interaction of radiation with matter: Absorption, scattering, and transmission. • Physical half-life, biological half-life, and effective half-life. • Radiation dosimetry: Exposure, absorbed dose, quality factor, and dose equivalent.	• In groups, use the animation to discuss the terms radiation dosimetry, exposure, absorbed dose, quality factor (relative biological effectiveness) and dose equivalent. • Through think-pair-share, discuss and distinguish between physical half-life, biological half-life and effective half-life. • In groups, discuss ionizing and non-ionizing radiations and differentiate between various imaging techniques.

• Explain the principles, operation, and applications of major imaging techniques. • Describe the basic principles of radiation therapy for cancer treatment. • Explain the biological effects of radiation and the concept of balanced risk.	• Apply principles of radiation protection.	• Develop curiosity and openness to advancements in medical imaging technology.	• Radiation safety and risk management. • Principles of sound in medical imaging. • Medical imaging technologies (radiographic, ultrasound, optical, nuclear medicine and magnetic resonance imaging). • Radiation therapy in medicine.	• In pairs, discuss the basics of radiation therapy for cancer treatment and present results. • In pairs, discuss safety precautions to be taken while handling radiations and concept of balanced risk. • Through group discussion, solved problems involving dosimetry calculations.
Links to other subjects: <i>Biology and Health Sciences (X-rays, CT scans, MRI). Chemistry (Radioactive isotopes)</i>				
Assessment criteria: <i>Learner correctly explains radiation and medical imaging principles, calculates doses and half-lives and effectively applies safety measures, and demonstrates ethical and responsible practice in healthcare settings.</i>				
Materials: <i>Animations, projector and computer.</i>				

TOPIC AREA: ELECTRONICS			SUB-TOPIC: ELECTRONIC DEVICES	
S6 PHYSICS	UNIT 9: INTRODUCTION TO ELECTRONICS		Nº of Periods: 20	
Key unit competence: Investigate how different electronic materials and devices work illustrating their structure and role in circuits.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Describe electronic materials. • Explain energy bands and electrical conductivity in materials. • Distinguish between intrinsic and extrinsic semiconductors. • Describe the structure and operation of a P-N junction diode, including depletion layer, forward and reverse biasing, and I-V characteristics.	• Draw the energy band diagrams for conductors, insulators, and semiconductors. • Demonstrate the forward and reverse biasing of a diode. • Sketch the I-V characteristics of biased diode.	• Develop precision and accuracy when building and testing electronic circuits. • Show curiosity and interest in exploring the behavior of electronic components.	• Introduction to electronic materials (insulators, semiconductors, conductors, energy bands and electrical conductivity). • Types of semiconductors (Intrinsic and extrinsic). • The P-N junction diode (depletion layer, forward and reverse biasing and I-V characteristics of a diode).	• Through think-pair-share, discuss and distinguish between conductors, insulators, and semiconductors in daily life. • In pairs, use AI based simulation to show electrical conductivity differences among materials using simple circuits. • Through group discussion, differentiate intrinsic from extrinsic semiconductors and discuss doping effect.

• Explain the working principle of Zener diodes and LEDs as examples of diodes. • Discuss the applications of diodes in rectifiers and filters. • Describe the structure and operation of a BJT transistor (emitter, base, collector). • Explain the applications of transistors as amplifiers and switches. • Describe the integrated circuit and its role. • Describe the function of basic logic gates: NOT, NOR, OR, AND, NAND.	• Build simple circuits using diodes and transistors for rectification, amplification, or switching. • Use logic gates to construct basic digital circuits and truth tables.	• Appreciate the role of electronics in modern technology and communication systems. • Promote collaboration, teamwork and safety awareness in practical circuit-building activities.	• Examples of diode (Zener & LED). • Applications of diodes (rectifiers and filters). • Transistor (BJT): Structure, emitter, base, collector, principle of operation). • Applications of transistors (amplifier & switch). • Integrated circuit. • Basic logic gates (NOT, NOR, OR, AND, NAND).	• Through guided inquiry, discuss ways of diode biasing, plot I-V characteristics and explain the formation and role of the depletion layer. • Through group discussion, describe a transistor, its structure and principle of operation. • Through group discussion, discuss and report on the applications of semiconductors. • Through guided inquiry, define a logic gate and build basic circuits using NOT, AND, OR, NAND, and NOR gates on breadboards or simulators. • Through group discussion, solve problems involving current flows in diodes and transistors.
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Links to other subjects: *Mathematics (equations, graphical analysis, Ratios and proportionality). Chemistry (electron energy levels and bonding principles, valency and periodic table).*

Assessment criteria: *Learner properly describes types of semiconductors; clearly explains the working principles of diodes and transistors; effectively plots and analyzes I-V characteristics, concisely discusses and reports on applications of diodes and transistors and correctly solves problems involving current flows (in diodes and transistors) and describe simple logic gates.*

Materials: *Breadboards and connecting wires, Resistors, capacitors, and inductors, Batteries or DC power supply, Switches and bulbs/LEDs, Multimeters (analog or digital), Crocodile clips and leads, Different types of diodes, Oscilloscope or simulation software, Transistors, Amplifier circuit boards, Signal generator, Speakers or buzzers, Logic gate ICs, Truth table charts, Digital circuit simulation software, LED indicators.*

TOPIC AREA: ASTROPHYSICS			SUB-TOPIC: EARTH AND SPACE	
S6 PHYSICS	UNIT 10: COSMOLOGY, GALAXIES AND EXPANSION OF UNIVERSE		Nº of Periods: 15	
Key unit competence: Analyze the structure and evolution of the Universe demonstrating the nature and properties of galaxies and stars.				
Learning objectives			Content	Learning activities
Knowledge and understanding	Skills	Attitudes and Values		
• Describe the importance of cosmology in understanding the universe. • Describe the structure and components of the Milky Way Galaxy. • Distinguish between the different types of galaxies (spiral, elliptical and irregular). • Explain the concept of galactic clusters and their role in large-scale structure of the universe.	• Interpret the expansion of the universe using Doppler shift and Hubble’s law. • Analyze stellar properties, distances, and masses based on radiation, spectra, and binary star data. • Interpret H–R diagrams to determine the characteristics and life stages of stars.	• Develop the curiosity about the origin, structure, and evolution of the universe. • Appreciate the vastness and complexity of the universe.	• Introduction to cosmology. • The structure of the Milky way Galaxy. • Types of Galaxies: spiral galaxies, elliptical galaxies, irregular galaxies. • Clusters of galaxies. • Expansion of the Universe (Doppler shift due to cosmic expansion and Hubble’s law).	• Through group discussion, use diagrams and simulations to explore the Milky Way Galaxy’s structure and visualize clusters. • In pairs, use simulations and animations to discuss and distinguish between spiral, elliptical and irregular galaxies.

• Explain the expansion of the universe using Doppler shift and Hubble's law. • Describe the main ideas and evidences of the Big Bang Theory. • Describe the structure and composition of the Sun (atmosphere and interior). • Explain the properties of stellar radiation, including brightness, magnitude, color, temperature, and spectra. • Describe methods used to determine stellar distances such as the parallax method. • Identify the relationship between stellar mass, luminosity, and temperature, and how binary stars help in determining stellar masses.	• Use data and graphs to apply Hubble's law for estimating the rate of expansion of the universe. • Calculate stellar distances using the parallax method. • Use models or simulations to illustrate the structure of the Milky Way and distribution of galaxies in clusters. • Solve problems involving brightness and magnitude scales in cosmology.	• Develop the environmental awareness inspired by understanding Earth's unique place in the cosmos. • Develop a critical thinking in analyzing astronomical data and interpreting cosmological models.	• Big Bang theory. • Stellar radiation and properties (Sun's atmosphere and interior structure, Brightness and magnitude scale, star temperature, color and spectra, spectral classification of stars). • Types of Stars and Hertzsprung-Russell Diagram (Main sequence, giants, supergiants, white dwarfs, Interpretation of H-R diagram).	• Through group discussion, use a sample of data to graph and analyze the plot for Hubble's Law. • In pairs, watch a simulation on Big Bang and explain the origin and evolution of the universe. • Through guided inquiry, conduct a balloon expansion experiment to model the universe's expansion • Through guided inquiry, plot an H-R diagram using data (temperature vs. luminosity) from sample stars and use this diagram to analyze the life cycle of a star.
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Links to other subjects: Geography and environmental sciences (climate change and seasons).

Assessment criteria: Learner accurately describes galaxies, expansion of the universe, types of stars and stellar radiation; properly plots and interprets Hertzsprung-Russell Diagram and correctly determines Stellar Distances and Masses and solves problems involving brightness and magnitude scales in cosmology.

Materials: Stellarium software, animation of the big bang, animations for spiral, elliptical and milky way galaxies.

6. REFERENCES

- Abbott, A. (1989). *Physics*. Chicago: Heinemann Educational Publishers.
- C., E., C., D., C., L., Lisowski, M., & J., J. (2006). *Science Explorer*. Mexico: Pearson Prentice Hall.
- Freedman, R. A., & Kaufmann III, W. J. (2008). *Stars and galaxies: Universe (3rd ed.)*. New York: W. H. Freeman and Company.
- Gibilisco, S. (2003). *Astronomy demystified*. New York: McGraw-Hill.
- Gibilisco, S. (2010). *Electronics demystified (2nd ed.)*. New York: McGraw-Hill.
- Hughes, P., & Mason, N. J. (2001). *Introduction to environmental physics: Planet Earth, life and climate*. London: Taylor & Francis.
- King, F. H. (1904). *A textbook of the physics of agriculture*. Madison, WI: Author.
- Kutner, M. L. (2003). *Astronomy: A physical perspective*. Cambridge: Cambridge University Press.
- NCDC, R. (2006). *Advanced level science curriculum*. Kigali: NCDC.
- Neil, F. C. (2009). *Discovering the universe: From the stars to the planets*. New York: W. H. Freeman and Company.
- Nelkon, M., & Parker, P. (1995). *Advanced level physics*. London: Heinemann.
- Palen, S. E. (2002). *Theory and problems of astronomy*. New York: McGraw-Hill.
- Richard, O. (2009). *Physics for Rwanda secondary school*. Kigali: Fountain Publishers.
- Tom, D. (2000). *Advanced physics*. London: Hodder Education.
- Valerio, F. (2003). *Exercises on environmental physics*. New York: Springer.
- W., J., & B., J. (2009). *Curriculum development: A guide to practice*. USA: Pearson Merrill Prentice Hall.
- Wysession, M., Frank, D., & Yancopoulos, S. (2004). *Physical science*. Boston, MA: Pearson Prentice Hall.

7. APPENDIX: SUBJECTS AND WEEKLY TIME ALOCATION FOR A'LEVEL

A. MATHEMATICS AND SCIENCE PATHWAY

Mathematics and Science Stream I

Subject	Number of periods (Period= 40 minutes)		
I. Major Core subjects: Examinable at National level	S4	S5	S6
1. Mathematics	7	7	7
2. Physics	7	7	7
3. Chemistry	7	7	7
4. Bilogy	7	7	7
II. Minor Core Subjects: Examinable at National level			
5. English	3	3	3
6. French	3	3	3
7. Kinyarwanda	3	3	3
Sub-total 1	37	37	37
III. General Subjects examinable at School level			
8. ICT	3	3	3
9. Entrepreneurship	3	3	3
10. General and Religious studies	3	3	3

11. Physical Education and Sports	2	2	2
Sub-total 2	11	11	11
III. Co-curricular activities (compulsory)	2	2	2
Total number of contact periods per week	50	50	50
Total number of contact hours/week	33.3	33.3	33.3
Total number of contact hours per year (39 weeks)	1300	1300	1300

Mathematics and Science Stream 2

Subject	Number of periods		
	(Period= 40 minutes)		
I. Major Core Subjects: Examinable at National Level	S4	S5	S6
1. Mathematics	7	7	7
2. Economics	7	7	7
3. Geography	7	7	7
4. Physics	7	7	7
II. Minor Core Subjects: Examinable at National Level			
5. English	3	3	3
6. French	3	3	3
7. Kinyarwanda	3	3	3
Sub-total 1	37	37	37

III. General Subjects examinable at School level			
8. ICT	3	3	3
9. Entrepreneurship	3	3	3
10. General and religious studies	3	3	3
11. Physical Education and Sports	2	2	2
Sub-total 2	11	11	11
III. Co-curricular activities (compulsory)	2	2	2
Total number of contact periods per week	50	50	50
Total number of contact hours/week	33.3	33.3	33.3
Total number of contact hours per year (39 weeks)	1300	1300	1300